Master's internship report

Entanglement measures and LOCC conversion criteria from Schur–Weyl duality

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Abstract

Computing entanglement measures of mixed states requires an optimization over all their pure-state decompositions, which is generally intractable. In this work, we introduce a method to optimize polynomials of the entanglement spectrum over mixed-state decompositions. We show how this approach can be used to approximate any entanglement measure with a convergent semidefinite programming hierarchy. Finally, we extend the approach to the LOCC convertibility problem, to certify when a pure state $|\psi\rangle$ cannot be converted into a mixed state ρ under local operations and classical communication (LOCC).

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Chapter 1

Introduction

Composite quantum systems are entangled when they cannot be described solely by their parts [1]. Operationally, entanglement is a resource for quantum communication [2, 3] and computation [4, 5] that cannot increase under local operations and classical communication (LOCC) between distant parties. In isolated two-particle quantum systems, described by pure states, entanglement and LOCC transformations are completely characterized by the Schmidt coefficients or the eigenvalues of the marginals [6, 7], also called the entanglement spectrum. Therefore, learning this spectrum is necessary and sufficient to understand the entanglement of the system and the possible entanglement transformations between states.

In practice, however, pure states become mixed due to decoherence with the environment [8, 9, 10]. When this occurs, the spectrum of the marginals alone is not informative of the entanglement between them, because a mixed state admits multiple descriptions as a probabilistic ensemble of pure states. Consequently, one needs to account for all such decompositions. This leads to an intractable optimization required to evaluate entanglement measures [11, 12, 13, 14], even when full tomography is available. Existing methods address each specific entanglement measure independently [15, 16, 17, 18, 19, 20, 21, 22]. Moreover, while lower bounds can be found for polynomial measures [23], much less is known for the more general case [24]. As a result, it is still a remarkable challenge to determine when entanglement transformations are possible with LOCC [25, 26, 27].

During the internship, we addressed these problems using the symmetries of multiple copies of a state. I was supervised by Albert Rico, whose research focuses on exploiting the symmetries of the Hilbert space to learn properties of quantum systems. The project took place at the University of Siegen in Germany, in the research group of Otfried Gühne, whose work mainly lies in entanglement theory. We started the project with the question of entanglement dimension and what Schur–Weyl duality can tell us about the Schmidt number. This led us to build a hierarchy of semidefinite programs to certify lower bounds on the Schmidt number of mixed states, and a method to bound the convex roof of the Vidal monotones. We built this hierarchy by extending the work of Géza Tóth and Otfried Gühne [23]. During this work, we also built a few libraries for tools we needed but could not find online, and we hope they will be useful to other researchers. As the project evolved, we extended this hierarchy to bound the convex roof of any pure-state polynomial entanglement measure, which we show can also be extended to non-polynomial measures such as the Vidal monotones or the von Neumann entropy. Finally, this led us to a hierarchy for LOCC non-convertibility from pure to mixed states. This report is based on our paper *Entanglement measures and LOCC conversion criteria from Schur–Weyl duality* [28], which is currently available on arXiv as arXiv:2610.37785.

Chapter 2

Background

2.1 Entanglement spectrum

Consider a bipartite pure state $|\psi\rangle_{AB} \in \mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B} \cong \mathbb{C}^D$ with $D = d_A d_B$. Such a state is separable if it can be written as $|\psi\rangle = |a\rangle \otimes |b\rangle$ and entangled otherwise. More generally, any pure bipartite state admits a *Schmidt decomposition*

$$|\psi\rangle_{AB} = \sum_{i=1}^{r} \sqrt{\alpha_i}\, |a_i\rangle \otimes |b_i\rangle\,, \tag{2.1}$$

where $\alpha_1 \geq \alpha_2 \geq \cdots \geq \alpha_r > 0$ and the sets $\{|a_i\rangle\}_i$ and $\{|b_i\rangle\}_i$ are orthonormal. The spectrum $\alpha = (\alpha_1, \ldots, \alpha_{d_A})$ of the reduced density matrix $\rho_A = \mathrm{tr}_B\, |\psi\rangle\langle\psi|$ (with $\alpha_i = 0$ for $i > r$) is called the *entanglement spectrum* of $|\psi\rangle_{AB}$, and the number r of nonzero coefficients is known as the *Schmidt rank*, denoted $\mathrm{SR}(\psi)$.

By Nielsen's theorem [6], a bipartite pure state $|\psi\rangle$ with entanglement spectrum $\alpha(\psi)$ can be converted by LOCC into another bipartite pure state $|\phi\rangle$ with entanglement spectrum $\alpha(\phi)$ if and only if $E_j(\psi) \geq E_j(\phi)$ for all $j = 1, \ldots, d_A - 1$, where

$$E_j(\psi) = 1 - \sum_{i=1}^{j} \alpha_i(\psi), \tag{2.2}$$

as defined in Ref. [29]. In this sense, the quantities $E_j(\psi)$ form a complete set of entanglement monotones, and all information about the entanglement of each state is encoded in the entanglement spectrum. By Ref. [7], every pure-state entanglement monotone is a concave, unitarily invariant function of ρ_A, which also implies that pure-state monotones are functions that only depend on the spectrum of ρ_A.

A mixed state ρ_{AB} admits infinitely many decompositions $\{p_i, |\psi_i\rangle\}$ such that

$$\rho_{AB} = \sum_i p_i\, |\psi_i\rangle\langle\psi_i|\,, \tag{2.3}$$

where $\sum_i p_i = 1$ and $p_i > 0$ for all i. We denote the set of all such decompositions by $\mathcal{D}(\rho)$.

We say that a mixed state is separable if it can be written as

$$\rho_{AB} = \sum_i p_i\ |a_i\rangle\langle a_i| \otimes |b_i\rangle\langle b_i|\,, \tag{2.4}$$

with $|a_i\rangle$ in Alice's system and $|b_i\rangle$ in Bob's. If a state cannot be written this way, we say it is entangled. A simple way to detect entanglement is the PPT (Peres–Horodecki) criterion [30, 31]:

every separable state satisfies $\rho_{AB}^{T_B} \succeq 0$, where T_B is the partial transpose on Bob's system. Hence, if $\rho_{AB}^{T_B}$ has at least one negative eigenvalue, the state is entangled. In $\mathbb{C}^2 \otimes \mathbb{C}^2$ and $\mathbb{C}^2 \otimes \mathbb{C}^3$, this test is sufficient to distinguish between separable and entangled states [31]. In higher dimensions, there exist PPT entangled states, which satisfy $\rho_{AB}^{T_B} \succeq 0$ even though they are entangled. These states belong to the set of states known as bound entangled states, and it is not known whether there exist bound entangled states which have a negative partial transpose. In general, deciding whether a state is separable is a hard problem.

A pure-state entanglement measure E can be extended to mixed states via the convex roof extension,

$$\widehat{E}(\rho) = \min_{\mathcal{D}(\rho)} \sum_i p_i E(\psi_i). \tag{2.5}$$

The Schmidt rank is extended to mixed states by the Schmidt number [32],

$$\mathrm{SN}(\rho) = \min_{\mathcal{D}(\rho)} \max_i \mathrm{SR}(\psi_i). \tag{2.6}$$

The convex roof of a faithful entanglement measure vanishes exactly on separable states, so computing it is at least as hard as the separability problem, and closed forms are only known for two-qubit states [15] or special families of states [16, 17]. These optimizations are generally intractable, even when the state is known, although known methods give upper bounds on these measures. Using Carathéodory's theorem, one can show that for any continuous function E of the entanglement spectrum, the minimum in Eq. (2.5) is attained by a decomposition into linearly independent terms $|\psi_i\rangle\langle\psi_i|$, so that the optimum needs at most r^2 terms, where $r = \mathrm{rank}(\rho)$ [12].

2.2 Schur–Weyl duality

A general k-qudit quantum state is a vector in the Hilbert space $(\mathbb{C}^d)^{\otimes k}$. Consider the two following groups acting on this space: the *symmetric group* S_k of permutations $\pi \in S_k$ of k elements, acting as

$$\pi(|\psi_1\rangle \otimes \ldots \otimes |\psi_k\rangle) = \left|\psi_{\pi^{-1}(1)}\right\rangle \otimes \ldots \otimes \left|\psi_{\pi^{-1}(k)}\right\rangle, \tag{2.7}$$

and the *general linear group* GL_d of invertible $d \times d$ matrices $g \in GL_d$, acting as

$$g(|\psi_1\rangle \otimes \ldots \otimes |\psi_k\rangle) = g\,|\psi_1\rangle \otimes \ldots \otimes g\,|\psi_k\rangle\,. \tag{2.8}$$

The actions of these two groups commute,

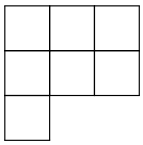
$$\pi\, g = g\, \pi. \tag{2.9}$$

Schur–Weyl duality [33] states that under the action of these two groups, the Hilbert space $(\mathbb{C}^d)^{\otimes k}$ decomposes as

$$(\mathbb{C}^d)^{\otimes k} \cong \bigoplus_{\substack{\lambda \vdash k \\ \ell(\lambda) \leq d}} \mathcal{S}_k^\lambda \otimes \mathcal{U}_d^\lambda \tag{2.10}$$

into irreducible subspaces $\mathcal{S}_k^\lambda$ and $\mathcal{U}_d^\lambda$, which carry the irreducible representations of S_k and GL_d respectively, indexed by partitions λ. A partition λ of k is a non-increasing vector of positive integers summing to k, and we represent it as a Young diagram: a diagram of stacked boxes

where the number of boxes in row i is equal to λ_i. For example, we write $\lambda = (3, 3, 1)$ in the following way



(2.11)

We denote the height of a Young diagram by $\ell(\lambda)$ and we write $\lambda \vdash k$ to denote that λ is a partition of k. A Young tableau is a filling of the boxes of a Young diagram with positive integers. It is called semi-standard if the entries are weakly increasing along each row and strictly increasing down each column, and it is called standard if it also contains each of the numbers $1, \ldots, k$ exactly once. For example,

semi-standard

1	1	2
2	2	4
3		

standard

1	2	6
3	5	7
4		

(2.12)

From these, we define the following constants: f^λ, the dimension of $\mathcal{S}_k^\lambda$, which is the number of standard Young tableaux of shape λ. The Schur polynomial s_λ is defined in Sec. 3.1, but we can note that $s_\lambda(1, 1, \ldots, 1)$ is the dimension of $\mathcal{U}_d^\lambda$, which is also equal to the number of semi-standard Young tableaux of shape λ with entries in $\{1, \ldots, d\}$. For $\lambda, \mu \vdash k$, the Kostka number $K_{\lambda\mu}$ is the number of semi-standard Young tableaux of shape λ with μ_1 ones, μ_2 twos, etc. From this definition, we can see that $K_{\lambda\mu} = 0$ if λ does not majorize μ, which implies that $K_{\lambda\mu} = 0$ if $\ell(\lambda) > \ell(\mu)$.

To each Young diagram λ we can associate an isotypic projector Π^λ projecting onto the subspace $\mathcal{S}_k^\lambda \otimes \mathcal{U}_d^\lambda$. This projector is given by

$$\Pi^\lambda = \frac{f^\lambda}{k!} \sum_{\pi \in S_k} \chi^\lambda(\pi)\, \pi, \tag{2.13}$$

where $\chi^\lambda(\pi)$ is the character of the cycle type of π in the irreducible representation λ. For example, in $(\mathbb{C}^d)^{\otimes 2}$, the Hilbert space decomposes into the symmetric space represented by $\lambda = (2)$ and the antisymmetric space $\lambda = (1, 1)$ with the following projectors

$$\Pi^{(2)} = \frac{\mathbb{1} + \mathrm{SWAP}}{2}, \quad \Pi^{(1,1)} = \frac{\mathbb{1} - \mathrm{SWAP}}{2}, \tag{2.14}$$

where SWAP is the swap gate.

Chapter 3

A complete hierarchy for polynomial measures

3.1 Polynomials of the entanglement spectrum

The spectrum of a reduced state ρ_A can be estimated via weak Schur sampling on k copies of ρ_A [34, 35]: the probability of measuring a Young diagram $\lambda \vdash k$ is given by

$$\mathrm{tr}\left(\Pi^\lambda \rho_A^{\otimes k}\right) = f^\lambda s_\lambda(\mathrm{spec}\,\rho_A). \tag{3.1}$$

It is known that the Schur polynomials s_λ with $\lambda \vdash k$ and $\ell(\lambda) \leq d_A$ form a basis of the homogeneous symmetric polynomials P of degree k of the spectrum of ρ_A [36], and they expand into the monomial basis in the following way:

$$s_\lambda(\mathrm{spec}\,\rho_A) = \sum_{\mu \vdash k} K_{\lambda\mu} m_\mu(\mathrm{spec}\,\rho_A), \tag{3.2}$$

where m_μ is the monomial symmetric polynomial

$$m_\mu(\alpha) = \sum_\beta \alpha_1^{\beta_1}\alpha_2^{\beta_2}\ldots\alpha_{d_A}^{\beta_{d_A}}, \tag{3.3}$$

where the sum runs over all *distinct* permutations β of $(\mu_1, \mu_2, \ldots, \mu_{\ell(\mu)}, 0, \ldots, 0) \in \mathbb{N}^{d_A}$. For example, with $\alpha = (\alpha_1, \alpha_2, \alpha_3)$,

$$\begin{aligned} s_{(2,1)}(\alpha) &= K_{(2,1)(3)} m_{(3)}(\alpha) + K_{(2,1)(2,1)} m_{(2,1)}(\alpha) + K_{(2,1)(1,1,1)} m_{(1,1,1)}(\alpha) \\ &= 0 \times m_{(3)}(\alpha) + 1 \times m_{(2,1)}(\alpha) + 2 \times m_{(1,1,1)}(\alpha) \\ &= \left(\alpha_1^2\alpha_2 + \alpha_1^2\alpha_3 + \alpha_2^2\alpha_1 + \alpha_2^2\alpha_3 + \alpha_3^2\alpha_1 + \alpha_3^2\alpha_2\right) + \left(2\alpha_1\alpha_2\alpha_3\right). \end{aligned} \tag{3.4}$$

Inverting this expansion shows how any monomial symmetric polynomial can be expressed in terms of Schur polynomials,

$$\begin{aligned} s_\lambda(\alpha) &= \sum_\mu K_{\lambda\mu} m_\mu(\alpha), \\ m_\mu(\alpha) &= \sum_\lambda (K^{-1})_{\mu\lambda} s_\lambda(\alpha). \end{aligned} \tag{3.5}$$

At $k = 2$, with $\alpha = \mathrm{spec}\,\rho_A$, we can express the linear entropy in terms of Schur polynomials,

$$S_L(|\psi\rangle) = 1 - \mathrm{tr}\left(\rho_A^2\right) = 1 - m_{(2)}(\alpha) = 2s_{(1,1)}(\alpha), \tag{3.6}$$

since $1 = (\sum_i \alpha_i)^2 = m_{(2)}(\alpha) + 2m_{(1,1)}(\alpha)$ and $s_{(1,1)} = m_{(1,1)}$. Hence, with the observable $2\Pi_A^{(1,1)} \otimes \mathbb{1}_B$, we get $S_L(|\psi\rangle) = \mathrm{tr}\left(2\Pi_A^{(1,1)} \rho_A^{\otimes 2}\right) = \langle\psi|^{\otimes 2}\, 2\Pi_A^{(1,1)} \otimes \mathbb{1}_B\, |\psi\rangle^{\otimes 2}$ for any state $|\psi\rangle$.

To generalize those polynomials to mixed states, we are interested in their convex roof. Since $\mathrm{tr}(\rho_A) = 1$ implies $(\sum_i \alpha_i)^{k-l} = 1$, any symmetric polynomial of degree $l < k$ can be lifted to a homogeneous symmetric polynomial of degree k by multiplying it by $(\sum_i \alpha_i)^{k-l}$ (Lemma A.1 in Appendix A). Then, for any tuple of coefficients $\vec{c} := (c_{\lambda_1}, \ldots, c_{\lambda_t})$, where t is the number of partitions of k, we define the following polynomial

$$P(\mathrm{spec}\,\rho_A) = \sum_{\lambda \vdash k} c_\lambda s_\lambda(\mathrm{spec}\,\rho_A), \tag{3.7}$$

with the associated observable

$$O_P = \sum_{\lambda \vdash k} \frac{c_\lambda}{f^\lambda} \Pi^\lambda, \tag{3.8}$$

such that $P(\mathrm{spec}\,\rho_A) = \mathrm{tr}\left(O_P^A \rho_A^{\otimes k}\right) = \langle\psi|^{\otimes k}\, O_P^A \otimes \mathbb{1}^B\, |\psi\rangle^{\otimes k}$. We get its convex roof as

$$\begin{aligned} \widehat{P}(\rho_{AB}) &= \min_{\mathcal{D}(\rho_{AB})} \sum_i p_i P(\mathrm{spec}\,\rho_{A,i}) = \min_{\mathcal{D}(\rho_{AB})} \sum_i p_i\, \mathrm{tr}\left(O_P^A \otimes \mathbb{1}^B\, |\psi_i\rangle\langle\psi_i|^{\otimes k}\right) \\ &= \min_{\mathcal{D}(\rho_{AB})} \mathrm{tr}\left(O_P^A \otimes \mathbb{1}^B \sum_i p_i\, |\psi_i\rangle\langle\psi_i|^{\otimes k}\right), \end{aligned} \tag{3.9}$$

where $\rho_{A,i} = \mathrm{tr}_B\, |\psi_i\rangle\langle\psi_i|$. $\widehat{P}$ is an entanglement monotone if P is concave [7]. For any P, we always have unitary invariance since P only depends on the spectrum.

3.2 The relaxation

The approach is an extension of the work from Ref. [23]. We use the fact that any polynomial entanglement measure can be bounded using optimizations over functions of several copies of a state.

Given a bipartite mixed state ρ_{AB} acting on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$, to any decomposition in $\mathcal{D}(\rho_{AB})$ we associate the state

$$\omega_k = \sum_i p_i\, |\psi_i\rangle\langle\psi_i|^{\otimes k} \tag{3.10}$$

acting on $(\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B})^{\otimes k}$ (see Eq. (3.9)).

The core idea is that a polynomial entanglement measure of degree not larger than k can be written as a linear function on ω_k. Then, optimizing over all decompositions amounts to optimizing over all ω_k with the constraint that ω_k is symmetric and fully separable with respect to the k copies. The separability constraint is difficult to implement, but it can be treated by a hierarchy of semidefinite approximations, similar to the separability criterion of symmetric extensions [37].

Explicitly, we can compute a sequence of lower bounds on Eq. (3.9) as $\widehat{P}(\rho) \geq P_{\mathrm{LB}_k}(\rho)$ with the following semidefinite programming (SDP) relaxation,

$$\begin{aligned} P_{\mathrm{LB}_k}(\rho) = \min_{\Omega_k}\ & \mathrm{tr}\left(O_P^A \otimes \mathbb{1}^B\, \Omega_k\right) \\ \text{s.t. }\ & \Omega_k \succeq 0, \\ & \Pi_{\mathrm{sym}}^{AB,k}\, \Omega_k\, \Pi_{\mathrm{sym}}^{AB,k} = \Omega_k, \\ & \mathrm{tr}_{2,\ldots,k}(\Omega_k) = \rho, \\ & \Omega_k^{T_l} \succeq 0 \quad \forall\, l \in \{1, \ldots, \lfloor \tfrac{k}{2} \rfloor\}. \end{aligned} \tag{3.11}$$

Here, O_P^A acts on $A_1, \ldots, A_k$. Let $m = \deg(P)$. When $m < k$, we can either evaluate O_P on $A_1, \ldots, A_m$ or lift P to a degree-k symmetric polynomial, as both give the same result by Lemma A.1. $\Pi_{\mathrm{sym}}^{AB,k}$ is the projector onto the k-copy symmetric subspace of $(\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B})^{\otimes k}$ across copies, and T_l denotes the partial transpose on the first l copies. The true ω_k is positive semidefinite, satisfies $\Pi_{\mathrm{sym}}^{AB,k} \omega_k \Pi_{\mathrm{sym}}^{AB,k} = \omega_k$ and has first marginal ρ, and since ω_k is separable between copies, $\omega_k^{T_l} \succeq 0$ for all $l \in \{1, \ldots, \lfloor \frac{k}{2} \rfloor\}$. Hence ω_k is in the feasible set of this SDP at every level k, so the SDP gives a lower bound on the convex roof, and we get a hierarchy of k-th order lower bounds.

As Ω_k is supported on the symmetric subspace $\mathrm{Sym}^k(\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}) \subset (\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B})^{\otimes k}$, we can exploit this symmetry to reduce the matrix size of Ω_k from D^k to $\dim \mathrm{Sym}^k(\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}) = \binom{k+D-1}{k}$. By projecting onto the range of ρ, we can further reduce this size to $\binom{k+r-1}{k}$, where $r = \operatorname{rank} \rho$. Moreover, the variable Ω_k can be taken real for real states. These symmetry reductions are explained in detail in Appendix C. For example, for a full-rank two-qutrit mixed state, $D = 9$ and $k = 3$, the matrix size of Ω_k is reduced from 729 to 165.

A partially transposed matrix is not Bose symmetric, so the PPT conditions have to be imposed on the larger space $\mathrm{Sym}^l(\mathbb{C}^r) \otimes \mathrm{Sym}^{k-l}(\mathbb{C}^r)$, which limits the gains of the symmetry reductions. In the example above, the PPT block at $l = 1$ has size $9 \times 45 = 405$, against 165 for Ω_k.

3.3 Convergence

Result 3.1. *The bound of Eq. (3.11) is saturated asymptotically,* $\lim_{k\to\infty} P_{\mathrm{LB}_k}(\rho) = \widehat{P}(\rho)$.

The idea of the proof is as follows: by the de Finetti theorem for Bose-symmetric states [38, 39], the m-copy marginal of any feasible Ω_k is asymptotically close to a mixture $\sum p_i \left|\phi_i\right\rangle\!\left\langle\phi_i\right|^{\otimes m}$ whose first system marginal is asymptotically close to ρ as k increases. By continuity of the convex roof, $P_{\mathrm{LB}_k} \geq \widehat{P}(\rho) - \delta_k$ where $\lim_{k\to\infty} \delta_k = 0$. Moreover, the bound is nondecreasing in k. Both proofs are given in Appendix B.

The convergence does not rely on the PPT constraints, but Ref. [40] shows that symmetric extensions converge faster with them, and numerically we get much better results with the PPT constraints.

3.4 Schmidt number

Naturally, using the isotypic polynomial basis allows us to detect the entanglement dimension, i.e. the Schmidt number [32], of a target mixed state. To see this, recall that $\Pi^\lambda \left|e_1\right\rangle \otimes \cdots \otimes \left|e_k\right\rangle = 0$ if fewer than $\ell(\lambda)$ distinct vectors of an orthonormal basis $\{\left|e_i\right\rangle\}$ appear in the product, and therefore $\operatorname{tr}\left(\Pi_A^\lambda \otimes \mathbb{1}_B \left|\psi\right\rangle\!\left\langle\psi\right|_{AB}^{\otimes k}\right) = 0$ if and only if $\ell(\lambda) > \mathrm{SR}(\psi_{AB})$ [41, 42]. Therefore, if $P_{\mathrm{LB}_k} \neq 0$ up to the solver precision, then no pure-state decomposition of ρ exists in which all states have Schmidt rank lower than $\ell(\lambda)$. This is formally described as follows:

Result 3.2. *If* $P_{\mathrm{LB}_k} > 0$ *is certified for* $O_P = \Pi^\lambda$ *with* $\lambda \vdash k$*, then* ρ *has Schmidt number at least* $\ell(\lambda)$.

As a simple example, we test this Schmidt number hierarchy on the state $\rho = p \left|\phi^+\right\rangle\!\left\langle\phi^+\right| + (1-p) \left|01\right\rangle\!\left\langle 01\right|$ acting on $\mathbb{C}^3 \otimes \mathbb{C}^3$, with $\left|\phi^+\right\rangle = \frac{1}{\sqrt{3}} \sum_{i=0}^{2} \left|ii\right\rangle$, studied in Ref. [43], which has Schmidt number 3 for $p > 0$, and we detect it from $p \approx 0.01$ at level $k = 3$ with $\lambda = (1,1,1)$. It is easy to see that ρ has Schmidt number three for any $p > 0$: every pure state in a decomposition of ρ lies in the span of $\left|\phi^+\right\rangle$ and $\left|01\right\rangle$, and since $p > 0$, every decomposition contains a state

$a\,|\phi^+\rangle + b\,|01\rangle$ with $a \neq 0$. Such a state has Schmidt rank three, since its coefficient matrix $\frac{a}{\sqrt{3}}\mathbb{1} + b\,|0\rangle\langle 1|$ is triangular with a nonzero diagonal.

3.5 Simplified relaxation

Since the observables we built only act on Alice's subsystem copies, we can simplify the SDP by extending copies of the marginals $\rho_{A,i}$ of Alice instead of adding full copies $|\psi_i\rangle$ at each level k. We then define the following hierarchy in terms of k, m, where k is the number of full state copies and m the degree of the polynomial.

For $m \geq k$, define the following extension acting on $(\mathbb{C}^{d_A})^{\otimes m} \otimes (\mathbb{C}^{d_B})^{\otimes k}$ (i.e., m copies of A and k copies of B),

$$\omega_{k,m} := \sum_i p_i\, |\psi_i\rangle\langle\psi_i|^{\otimes k} \otimes \rho_{A,i}^{\otimes(m-k)} \tag{3.12}$$

with $\rho_{A,i} = \mathrm{tr}_B(|\psi_i\rangle\langle\psi_i|)$. We relax this operator with a positive semidefinite variable $\Phi_{k,m}$, and denote $\Phi_k := \mathrm{tr}_{A_{k+1},\ldots,A_m}(\Phi_{k,m})$. Then we can lower bound $\widehat{P}(\rho)$ as $P_{\mathrm{LB}_{k,m}} \leq \widehat{P}(\rho)$, where

$$\begin{aligned} P_{\mathrm{LB}_{k,m}} := \min_{\Phi_{k,m}} \ & \mathrm{tr}\Big(O_P^A \otimes \mathbb{1}^B\ \Phi_{k,m}\Big) \\ \text{s.t.}\ \ & \Phi_{k,m} \succeq 0, \\ & \Pi_{\mathrm{sym}}^{AB,k}\, \Phi_k\, \Pi_{\mathrm{sym}}^{AB,k} = \Phi_k, \\ & \mathrm{tr}_{A_2,\ldots,A_m,B_2,\ldots,B_k}(\Phi_{k,m}) = \rho_{AB}, \\ & \mathrm{tr}_{B_1,\ldots,B_k}(\Phi_{k,m}) \text{ is permutation invariant}, \\ & \Phi_{k,m}^{T_S} \succeq 0 \quad \forall\, S. \end{aligned} \tag{3.13}$$

Here O_P^A is the observable of Eq. (3.8) for a polynomial of degree m, acting on $A_1, \ldots, A_m$, and T_S denotes the partial transpose on a set S of registers, where transposing a full copy acts on both A_i and B_i. By permutation invariance, it suffices to impose the partial transpose conditions on a small set of representatives (Appendix B.3).

For $k \geq \deg P$, the marginal Φ_k of any feasible $\Phi_{k,m}$ is feasible for Eq. (3.11) at level k, so that $P_{\mathrm{LB}_k} \leq P_{\mathrm{LB}_{k,m}}$ and this hierarchy is also complete in k. In fact, notice that for $k = m$ the hierarchy of Eq. (3.13) reduces to Eq. (3.11). For any feasible Ω_m of Eq. (3.11) at level m, we also have that $\mathrm{tr}_{B_{k+1}\ldots B_m}(\Omega_m)$ is feasible for Eq. (3.13) at level (k, m), so that $P_{\mathrm{LB}_{k,m}} \leq P_{\mathrm{LB}_m}$.

This simplified relaxation allows us to access higher degree polynomials without paying the full cost of m copies of $|\psi_i\rangle\langle\psi_i|$, though the full hierarchy gives better results for polynomials of the same degree. The parameters m and k can be used for a tradeoff between performance and efficiency. In terms of memory usage, this hierarchy is not necessarily less costly at level (k_1, m_1) with $k_1 < m_1$ than it is at level (k_2, m_2) with $k_2 = m_2$. This depends on the rank of the target state ρ and its marginals (see Appendix C.5). Typically, level (k_1, m_1) is less costly for full-rank states while (k_2, m_2) is less costly for low-rank ones.

Observation 3.3. *For full-rank states, the simplified hierarchy achieves tighter bounds at the same computational cost.*

We show the memory cost of the hierarchy at different levels for states of different ranks in Fig. C.1 (Appendix C.6).

Chapter 4

Approximating the optimal decomposition

The goal of this chapter is to show how to extract the decomposition $\{p_i, |\psi_i\rangle\}$ from the optimal solution Ω_k found in the SDP, which gives us an upper bound to complement the lower bounds from the SDP. Notice that if the relaxation is tight, we have $\Omega_k = \omega_k = \sum_i p_i \, |\psi_i\rangle\langle\psi_i|^{\otimes k}$. This is in general not the case, and therefore we write $\Omega_k = \omega_k + \Delta$, where Δ accounts for the gap of the relaxation. For $k \geq 3$, the decomposition of ω_k into a sum of tensor powers is unique up to phases when the operators $|\psi_i\rangle\langle\psi_i|$ are linearly independent [44]. When the relaxation is tight and $\Omega_k = \omega_k$, we can then retrieve it exactly with Jennrich's algorithm [45, 46]. Since Ω_k is in general not of this form, we instead use the PARAFAC (or CANDECOMP) algorithm [45, 47] from the TensorLy package [48], initialized with Jennrich's algorithm, to find a decomposition $\{p_j, |\psi_j\rangle\}$ that locally minimizes the Frobenius norm

$$\Theta_k := \Big\| \Omega_k - \sum_j p_j \, |\psi_j\rangle\langle\psi_j|^{\otimes k} \Big\|_F, \tag{4.1}$$

with $\|X\|_F = \sqrt{\sum_{i,j} |X_{ij}|^2}$.

4.1 Heuristic extraction

To reduce the cost, we first project Ω_k onto the support of ρ, of dimension $r = \operatorname{rank} \rho$. PARAFAC extracts real vectors, while we want the projectors $|\psi_i\rangle\langle\psi_i|$. We therefore expand Ω_k in a real basis of Hermitian operators on $\mathbb{C}^r$, in which every pure state is a real unit vector x with a fixed first entry, so that ω_k becomes the real symmetric tensor $\sum_i p_i \, x^{(i)\otimes k}$ (Appendix D.1). We run PARAFAC on this tensor with as many terms as the rank of the projected Ω_k, and recover each pure state $|\psi_i\rangle$ as the top eigenvector of the corresponding Hermitian operator. For $k = 2$, the tensor is a matrix, whose decomposition is not unique, so the extraction needs $k \geq 3$.

The extraction improved considerably with a rank-reduction step before it: a second SDP selects, among the near-optimal solutions, a low-rank extreme point instead of a mixture of several optimal solutions (Appendix D.2).

4.2 Repair step

The extracted terms decompose the state $\sigma = \sum_j p_j \, |\psi_j\rangle\langle\psi_j|$, which is in general different from, but close to, ρ. We turn them into an exact decomposition of ρ by defining

$$|\varphi_j\rangle = \rho^{1/2} \sigma^{-1/2} \sqrt{p_j} \, |\psi_j\rangle \, , \tag{4.2}$$

where $\sigma^{-1/2}$ is taken on the support of σ. If $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma$, then

$$\sum_j |\varphi_j\rangle\langle\varphi_j| = \rho^{1/2}\sigma^{-1/2}\Big(\sum_j p_j\,|\psi_j\rangle\langle\psi_j|\Big)\sigma^{-1/2}\rho^{1/2} = \rho^{1/2}\sigma^{-1/2}\sigma\sigma^{-1/2}\rho^{1/2} \tag{4.3}$$

$$= \rho^{1/2}\Big(\sum_{e_{\sigma,i}>0} e_{\sigma,i}^{-\frac{1}{2}}\; e_{\sigma,i}\; e_{\sigma,i}^{-\frac{1}{2}}\,|\psi_{\sigma,i}\rangle\langle\psi_{\sigma,i}|\Big)\rho^{1/2} = \rho^{1/2}\Pi_\sigma\rho^{1/2} = \rho, \tag{4.4}$$

where $e_{\sigma,i}$ and $|\psi_{\sigma,i}\rangle$ are the eigenvalues and eigenvectors of σ and Π_σ is the projector onto the support of σ. The last equality holds because $\operatorname{supp}\rho^{1/2} = \operatorname{supp}\rho \subseteq \operatorname{supp}\sigma$, so that $\Pi_\sigma\rho^{1/2} = \rho^{1/2}$. Hence $\{q_j, |\phi_j\rangle\}$ with $q_j = \|\varphi_j\|^2$ and $|\phi_j\rangle = |\varphi_j\rangle / \|\varphi_j\|$ is a decomposition of ρ and gives a valid upper bound on the convex roof.

4.3 Upper bounds by gradient descent

We also compare our method to a more standard way of obtaining upper bounds, which is gradient descent parametrized by the HJW theorem [49]. We start with M vectors, $\left|\widetilde{\psi}_i\right\rangle = \sqrt{e_{\rho,i}}\,|\psi_{\rho,i}\rangle$ for $1 \le i \le r$ and $\left|\widetilde{\psi}_i\right\rangle = 0$ for $r < i \le M$, where $e_{\rho,i}$ and $|\psi_{\rho,i}\rangle$ are the eigenvalues and eigenvectors of ρ. For any $M \times M$ unitary U, the vectors $\left|\widetilde{\phi}_j\right\rangle = \sum_i U_{ij}\left|\widetilde{\psi}_i\right\rangle$ give us a decomposition of ρ:

$$\sum_j \left|\widetilde{\phi}_j\right\rangle\!\left\langle\widetilde{\phi}_j\right| = \sum_j \left(\sum_i U_{ij}\left|\widetilde{\psi}_i\right\rangle\right)\left(\sum_{i'} U^\dagger_{ji'}\left\langle\widetilde{\psi}_{i'}\right|\right) = \sum_{ii'}\sum_j U_{ij}U^\dagger_{ji'}\left|\widetilde{\psi}_i\right\rangle\left\langle\widetilde{\psi}_{i'}\right|$$

$$= \sum_{ii'}(UU^\dagger)_{ii'}\left|\widetilde{\psi}_i\right\rangle\left\langle\widetilde{\psi}_{i'}\right| = \sum_{ii'}\delta_{ii'}\left|\widetilde{\psi}_i\right\rangle\left\langle\widetilde{\psi}_{i'}\right| = \sum_i \left|\widetilde{\psi}_i\right\rangle\!\left\langle\widetilde{\psi}_i\right| = \rho.$$

Normalizing, $p_j = \left\langle\widetilde{\phi}_j\middle|\widetilde{\phi}_j\right\rangle$ and $|\phi_j\rangle = \left|\widetilde{\phi}_j\right\rangle / \sqrt{p_j}$ give $\rho = \sum_j p_j\,|\phi_j\rangle\langle\phi_j|$. The converse also holds: any decomposition of ρ into M terms can be written in this way [49]. As mentioned in Sec. 2.1, we only need $M = r^2$, since there exists a decomposition into at most r^2 terms for which the minimum is reached. Therefore every unitary provides a decomposition of ρ and hence an upper bound on the convex roof. As in Ref. [50], we minimize this upper bound by gradient descent over the unitaries, with several random starting points, and keep the best value.

4.4 Convergence of the extraction

Result 4.1. *Let P be a symmetric polynomial in pure-state Schmidt coefficients. It is possible to extract the decomposition of Ω_k optimizing P. By doing so, we obtain an upper bound $P_{\mathrm{UB}_k}(\rho) \ge \widehat{P}(\rho)$ which converges, $\lim_{k\to\infty} P_{\mathrm{UB}_k}(\rho) = \widehat{P}(\rho)$.*

The proof (Appendix D.3) combines three ingredients: the rank reduction selects an extreme optimal decomposition; extreme optimal decompositions consist of linearly independent projectors, so Jennrich's algorithm [45, 46] recovers them uniquely and continuously; and the repair step of Eq. (4.2) turns the result into an exact decomposition of ρ. In practice, we cannot run the SDP at k large enough to use Jennrich's algorithm and the extracted decomposition is always a heuristic one from PARAFAC.

We show the bounds on the convex roof of the linear entropy $S_L(\psi) = 1 - \operatorname{tr}\rho_A^2$ in Fig. 4.1.

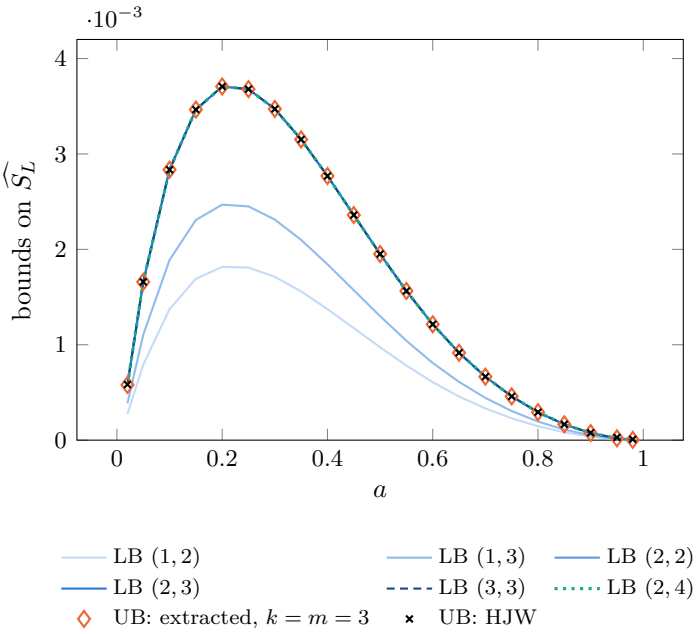


Figure 4.1: Convex roof of the linear entropy $S_L = 1 - \operatorname{tr} \rho_A^2$ for the Horodecki 2×4 family. Lower bounds are shown at different levels of the hierarchy. The markers show the upper bounds extracted from our method and HJW gradient descent.

Chapter 5

Non-polynomial measures via linear programming

Now that we have defined the tools we use to bound polynomial entanglement measures, we want to extend the method to non-polynomial measures. Our goal is therefore to find the polynomial that best approximates the target measure. Namely, we aim to find

$$\widetilde{F}_{\mathrm{LB}_k}(\rho) = \sup_{\substack{P\leq F \\ \deg P\leq k}} \min_{\Omega_k} \operatorname{tr}\left(\left(O_P^A \otimes \mathbb{1}^B\right)\Omega_k\right), \tag{5.1}$$

the optimization over symmetric polynomials of degree at most k that lower bound F on all spectra. Since the hierarchy of Eq. (3.11) is complete and any continuous function can be approximated by a polynomial arbitrarily well on a compact set, we can use the polynomial hierarchy to get asymptotic bounds on non-polynomial entanglement measures.

In practice, we cannot run the SDP at the levels k needed to reach the asymptotic regime, so we introduce a method based on linear programming to get better bounds at lower levels. Using that all symmetric polynomials can be written in the (discrete) basis of Schur polynomials, we will show that finding the best polynomial that lower bounds any entanglement measure is a problem that can be cast as a linear program (LP) over the coefficients in this basis. This allows us to dynamically find the best lower-bounding polynomial for a given state while running the SDP, since linear programs are a special case of semidefinite programs.

The proof is constructive. First, we select multiple spectra $\alpha = (\alpha_1, \ldots, \alpha_{d_A})$ at which we compute all the Schur polynomials $s_\lambda(\alpha)$ of degree k and the input function $F(\alpha)$. These are points of the simplex of possible spectra α, and define a grid $G = \{\alpha^g\}$ discretizing this simplex of possible spectra (Fig. 5.1). Here the superindex α^g is chosen to distinguish the grid point g from the subindex j denoting each corresponding eigenvalue α_j^g. We then require $F(\alpha^g) \geq \sum_\lambda c_\lambda s_\lambda(\alpha^g)$ at every spectrum $\alpha^g \in G$. By symmetry of the Schur polynomials, we only need to compute the points where $\alpha_i \geq \alpha_j$ for $i < j$ for each spectrum α. Thus, we are left with the following optimization problem,

$$F_{\mathrm{LB}_k}(\rho) = \max_{\substack{P(\alpha^g)\leq F(\alpha^g)\,\forall g \\ \deg P\leq k}} \min_{\Omega_k} \operatorname{tr}\left(\left(O_P^A \otimes \mathbb{1}^B\right)\Omega_k\right). \tag{5.2}$$

In order to solve this optimization, we need to address two challenges. First, a certain error ϵ_k is inherent to this grid procedure, in the sense that $|F_{\mathrm{LB}_k} - \widetilde{F}_{\mathrm{LB}_k}| \leq \epsilon_k$ (Fig. 5.1). Since we can retrieve the polynomial as output from the SDP, we can compute this error numerically and subtract it from the output to obtain a lower bound, namely $\widetilde{F}_{\mathrm{LB}_k} \geq F_{\mathrm{LB}_k} - \epsilon_k$. Second, this is in principle a min-max optimization problem. However, since the LP is feasible and has a

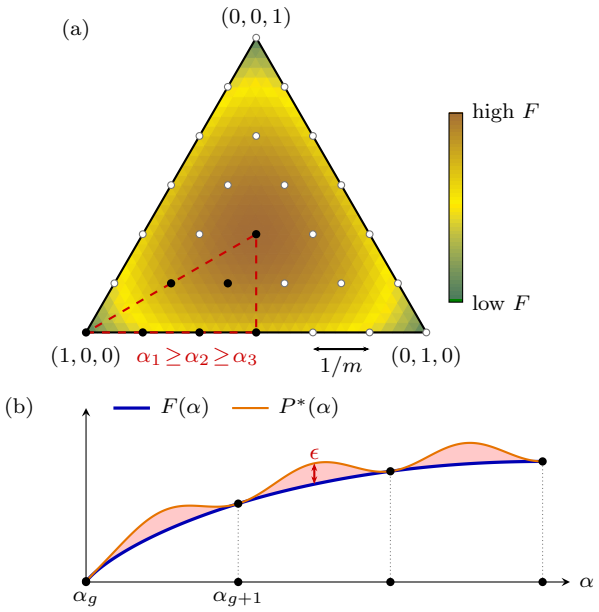


Figure 5.1: Grid discretization of the simplex of spectra for $d_A = 3$. (a) Different values of $F(\alpha)$ are denoted by different colors. The dots denote the grid G, separated by a certain distance $1/m$. Since entanglement measures are symmetric functions on the spectrum, only the points with $\alpha_1 \geq \alpha_2 \geq \alpha_3$ (dashed region, filled dots) need to be imposed. (b) Cut along a one-parameter family of spectra. Although $P^* \leq F$ holds at all grid points, it is possible that $P^* > F$ between any two points by a value ϵ. This value decreases as we refine the grid.

bounded solution, strong duality holds, and we can turn this problem into a single minimization problem by taking the dual of the LP. Details are given in Appendix E.1.

We can note that as the number of points on the grid increases, the grid error ϵ between grid points will tend to decrease, giving us better bounds.

Result 5.1. *We obtain a complete hierarchy to estimate any entanglement measure on mixed states, in the following sense:*

1. *Let F be a continuous symmetric function of the entanglement spectrum, and let $\widehat{F}(\rho)$ be its convex roof (2.5). Then $\widetilde{F}_{\mathrm{LB}_k}(\rho) \leq \widehat{F}(\rho)$ at all levels k.*
2. *The bound is non-decreasing in k and convergent, $\lim_{k\to\infty} \widetilde{F}_{\mathrm{LB}_k}(\rho) = \widehat{F}(\rho)$.*
3. *With a finite grid G of spectra $\alpha^g \in G$, $F_{\mathrm{LB}_k}(\rho) - \epsilon \leq \widehat{F}(\rho)$, where $\epsilon = \max_\alpha (P^*(\alpha) - F(\alpha))$, taken over all spectra, is the grid error of the polynomial P^* returned by the solver.*

The proofs of all statements above are given in Appendix E.2. We apply this method to estimate the convex roof of the von Neumann entropy of entanglement,

$$S(|\psi_{AB}\rangle) = -\operatorname{tr}(\rho_A \log_2 \rho_A), \tag{5.3}$$

where $\rho_A = \operatorname{tr}_B |\psi_{AB}\rangle\langle\psi_{AB}|$, extended to mixed states via convex roof, which we denote as $E_F = \widehat{S}$. This is known as the entanglement of formation [11], and it is well known to be difficult to compute [20]. To benchmark the method introduced, we use it to estimate $\widehat{S}(\rho_{AB})$ for several families of mixed states ρ_{AB} of different local dimensions, including the main known bound entangled states. The results are given in Fig. 5.2, and in more detail in Fig. G.1 in Appendix G.

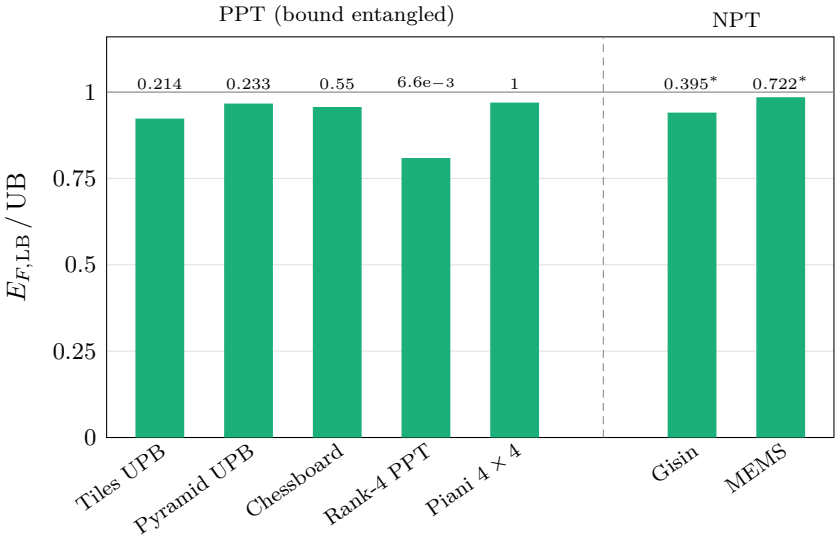


Figure 5.2: Best lower bound on the entanglement of formation E_F for different PPT and NPT states, from the SDP with embedded LP at level $(k, m) = (2, 4)$, divided by the best upper bound UB printed above each state. For two qubits, UB is the exact convex roof (marked *). All levels of the hierarchy and both upper bounds, for more states, are shown in Fig. G.1.

Chapter 6

LOCC convertibility

Finally, let us demonstrate how the presented approach allows us to determine whether certain pairs of states can be transformed by means of LOCC. To date, and to the best of our knowledge, the only criteria known for this problem are restricted to qubits and very special families of states [26, 25].

We can extend the method above to tackle the problem of LOCC convertibility from pure to mixed states, which is in principle an optimization over multiple non-polynomial functions. According to Ref. [51], a pure state $|\psi\rangle$ can be converted into an ensemble $\{p_i, |\psi_i\rangle\}$ by LOCC if and only if

$$E_j(\psi) \geq \sum_i p_i E_j(\psi_i) \qquad \text{for all } j = 1, \ldots, d_A - 1, \tag{6.1}$$

where E_j are the Vidal monotones (2.2). Hence, $|\psi\rangle$ can be converted into a mixed state ρ if and only if some decomposition of ρ satisfies all these inequalities simultaneously.

This is not straightforward to verify, since the decompositions of ρ minimizing each monotone E_j might be different. Assuming we don't use a grid of spectra, let $\mathcal{P}_j$ be the set of polynomials P_j such that $P_j(\alpha) \leq E_j(\alpha)$ on all spectra. We want to solve the following optimization problem

$$\widetilde{s}_k = \max_t \max_{P_j \in \mathcal{P}_j} \min_{\Omega_k} \sum_j t_j \Big(\operatorname{tr}\left((O_{P_j} \otimes \mathbb{1}^B)\Omega_k\right) - E_j(\psi)\Big), \tag{6.2}$$

where t is a probability vector. If $\widetilde{s}_k > 0$, then there is at least one Vidal monotone that is not satisfied and LOCC conversion is not possible.

We convert this into an SDP by combining the LPs of Chapter 5 for all the E_j into a single SDP and enforcing the polynomial constraints on a grid of spectra. This gives us the following optimization, where $\mathcal{P}_j^G$ is the set of polynomials satisfying the grid constraint:

$$s_k = \min_{\Omega_k} \max_t \max_{P_j \in \mathcal{P}_j^G} \sum_j t_j \Big(\operatorname{tr}\left((O_{P_j} \otimes \mathbb{1}^B)\Omega_k\right) - E_j(\psi)\Big). \tag{6.3}$$

We can swap the max and min since the objective is bilinear and the set of Ω_k is convex and compact (Sion's theorem [52]). The complete SDP is given in Appendix F.1. From its dual, we can extract the weights t_j^* and polynomials $Q_j^* = t_j^* P_j^*$ that satisfy $Q_j^* \leq t_j^* E_j$ on the grid.

As shown in Chapter 4, we can extract the decomposition from the result of the SDP. This leads us to a certificate that certain pure-to-mixed LOCC transformations are possible via Eq. (6.1). This can be summarized as follows.

Result 6.1. *By extracting a decomposition $\{p_i, |\psi_i\rangle\}$, we can verify whether it satisfies Eq. (6.1) for a certain input state $|\psi\rangle$, and therefore determine whether they can be transformed with LOCC.*

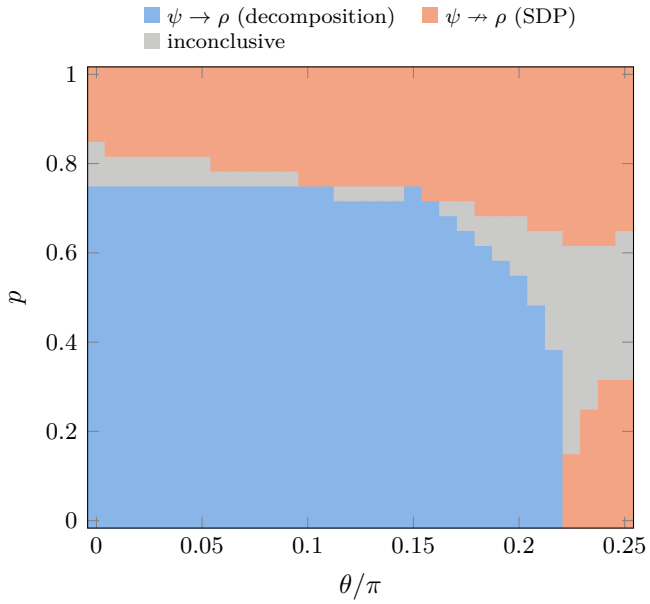


Figure 6.1: LOCC convertibility from the pure state $|\psi\rangle$ with entanglement spectrum $(0.6, 0.3, 0.1)$ to the two-qutrit states $\rho(p,\theta) = p\,|\Phi_3\rangle\langle\Phi_3| + (1-p)\,|\chi_\theta\rangle\langle\chi_\theta|$, with $|\Phi_3\rangle$ maximally entangled and $|\chi_\theta\rangle = \cos\theta\,|00\rangle + \sin\theta\,|11\rangle$, from the SDP of Eq. (6.3) at level $(k,m) = (3,4)$.

However, unlike for the bounds for polynomial and non-polynomial measures, it may not always be possible to extract the correct decomposition. This comes from the fact that there may not exist a decomposition of ρ into linearly independent terms that satisfies the criterion of Eq. (6.1), which makes it very difficult for the algorithms of Chapter 4 to find the underlying decomposition. Instead, a similar strategy allows us to obtain a complete no-go criterion to discard entanglement transformations.

Result 6.2. *We obtain a complete hierarchy for LOCC non-convertibility in the following sense:*

1. *Let* $\epsilon_j = \max_\alpha \left(Q_j^*(\alpha) - t_j^* E_j(\alpha)\right)$*, taken over all spectra, be the grid errors of the polynomials returned by the solver. If* $s_k > \sum_j \epsilon_j$ *for some* k*, then* $|\psi\rangle$ *cannot be converted into* ρ *by LOCC.*

2. *In the limit where the grid* G *is continuous and covers all possible spectra* α*, for any* $|\psi\rangle$ *and* ρ*, if* $|\psi\rangle \overset{LOCC}{\nrightarrow} \rho$*, there is a finite level* k *at which* $\widetilde{s}_k > 0$*.*

The proof is given in Appendix F. Notice that although the grid will always be discrete in practical computations, by continuity the second item suggests that finer grids lead to tighter criteria. Numerically, we verify that this is the case for all states tested.

Chapter 7

Libraries

7.1 `schur-weyl`

When implementing the method, we found very few online resources providing the tools from Schur–Weyl duality that we needed for our code. At the start of the project we used SageMath [53], which provides the character tables needed to build the isotypic projectors of Eq. (2.13), which are hard to compute in general, as well as Schur polynomials. However, SageMath is heavy (about 3 GB) and still required a lot of code to build the tools we needed. We therefore built a small Python package, `schur-weyl` [54], that provides the tools used in this project.

The content of the package is listed in Table 7.1.

Module	Contents
`young_diagrams`	Partitions and the relevant helpers
`dimensions`	$f^\lambda = \dim \mathcal{S}_k^\lambda$ and $\dim \mathcal{U}_d^\lambda$
`tableaux`	Standard and semi-standard Young tableaux, Kostka numbers
`symmetric_group`	Permutations, composition of permutations, inverse, cycle type of a permutation, conjugacy class sizes
`character`	Characters χ^λ of S_k
`symmetric_functions`	Schur polynomials, power sums, monomial symmetric polynomials and their expansion in the Schur basis
`isotypic`	Isotypic projectors Π^λ
`sw_measure`	Weak Schur sampling distribution $f^\lambda s_\lambda(\operatorname{spec}\rho)$
`schur_transform`	Schur transform: a real orthogonal matrix block-diagonalising the actions of S_k and GL_d simultaneously

Table 7.1: Modules of `schur-weyl` [54].

7.2 `bound-entangled`

As the project started, our first goal was to test the hierarchy above to measure the Schmidt number and see whether it is possible to do better than projecting on the antisymmetric projector. This is not possible by Lemma A.1.

To test this, we needed access to families of states on which we could run the SDP. At the time, the SDP was very costly, at about 15 GB of memory at level $k = 3$ for two-qutrit states, which led us to mostly test it on the separability problem for bound entangled states, since PPT

entangled states are not detected by the PPT criterion. This led us to build a repository of families of bound entangled states, `bound-entangled` [55].

The states it contains are listed in Table 7.2.

Space	States
$\mathbb{C}^2 \otimes \mathbb{C}^4$	Horodecki 2×4 state
$\mathbb{C}^3 \otimes \mathbb{C}^3$	Horodecki 3×3 state, chessboard states, cross-hatch grid state, Vértesi–Brunner state violating a Bell inequality, steerable state disproving the stronger Peres conjecture, Tiles and Pyramid UPB states, six-parameter family of UPBs generalising them
$\mathbb{C}^4 \otimes \mathbb{C}^4$	Piani state
$\mathbb{C}^5 \otimes \mathbb{C}^5$	Grid state with Schmidt number 3
$\mathbb{C}^d \otimes \mathbb{C}^d$	Yu–Oh states, GenTiles1 UPB states, private-singlet states of Badziąg et al., second family of PPT singlets, Horodecki state generalised to $\mathbb{C}^2 \otimes \mathbb{C}^d$, Breuer states
$\mathbb{C}^m \otimes \mathbb{C}^n$	Grid states and their generalisation to hypergraphs, GenTiles2 UPB states
Multipartite	Smolin state, generalised Smolin states, quasi-DS states

Table 7.2: States implemented in `bound-entangled` [55], by Hilbert space. UPB: unextendible product basis.

Chapter 8

Conclusions

By using a unitarily invariant basis of projectors, which lets us compute polynomials of the entanglement spectrum on pure states, we have introduced an SDP formulation to bound any entanglement measure of bipartite mixed states. This method was extended to the problem of LOCC convertibility with a hierarchy of certificates of non-convertibility and a heuristic approach towards certifying convertibility. A natural direction for future work would be to see how this method can be extended to the multipartite setting. I would also like to see how we could further optimize the SDP, either by increasing the convergence rate by adding more separability constraints along the systems of Ω_k such as realignment criterion constraints, or by improving the symmetry reductions and the choice of grid points for the LP.

Contributions. This report is based on the paper [28], written with Albert Rico and Otfried Gühne, and a significant part of the text is adapted from it. I wrote the main body of the paper as well as the appendices with some edits from my supervisors. The conclusion and introduction of the paper were mainly written by my co-authors. I also developed the software libraries.

AI statement. The ideas, methodology, proof strategies and the project scope were designed by the authors. I used AI coding tools (Claude Opus 4.8–5.5) to write and debug part of the code, including the software libraries, to run and plot numerical experiments and to check proof calculations. All results were validated by me.

Acknowledgements. Thank you to Albert Rico and Otfried Gühne for welcoming me into the group and for all their help during the internship. Thank you to everyone from the University of Siegen for the great time I had.

Appendix A

Representation theory and lifting

A.1 Lifting to higher degree

Lemma A.1. *Let P be a homogeneous symmetric polynomial of degree m with $m < k$. Let P_k be its lift to degree k such that $P_k(\alpha) = P(\alpha)\,(\sum_i \alpha_i)^{k-m} = P(\alpha)$ where α is the spectrum of a state ρ_A. Then, for every Ω_k acting on $(\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B})^{\otimes k}$, which is permutation invariant across all k copies of $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$,*

$$\mathrm{tr}\Big(O_P^{A_1,\ldots,A_m} \otimes \mathbb{1}^{A_{m+1},\ldots,A_k} \otimes \mathbb{1}^B \Omega_k\Big) = \mathrm{tr}\Big(O_{P_k}^{A} \otimes \mathbb{1}^B \Omega_k\Big). \tag{A.1}$$

Proof. Let $\mathcal{T} = \frac{1}{k!}\sum_{\pi\in S_k} \pi\, O_P \otimes \mathbb{1}^{\otimes(k-m)}\, \pi^\dagger$ be the twirl of $O_P \otimes \mathbb{1}^{\otimes(k-m)}$ over S_k. Obviously $\mathcal{T}$ commutes with every permutation in S_k, and, since it is a linear combination of permutations, we can write it as a sum of isotypic projectors $\mathcal{T} = \sum_{\lambda\vdash k} a_\lambda \Pi^\lambda$ for some coefficients a_λ.

For any ρ_A with spectrum α, since $\rho_A^{\otimes k}$ is permutation invariant,

$$\mathrm{tr}\Big(\mathcal{T}\rho_A^{\otimes k}\Big) = \mathrm{tr}\Big(\frac{1}{k!}\sum_{\pi\in S_k} \pi\, O_P \otimes \mathbb{1}\, \pi^\dagger \rho_A^{\otimes k}\Big) = \mathrm{tr}\Big(O_P \otimes \mathbb{1}\, \rho_A^{\otimes k}\Big). \tag{A.2}$$

Thus $\mathcal{T}$ and O_{P_k} have the same expectation value $P_k(\alpha)$ on every $\rho_A^{\otimes k}$, and since both are combinations of isotypic projectors and the Schur polynomials are linearly independent, $\mathcal{T} = O_{P_k}$. Finally,

$$\mathrm{tr}\Big(O_P^{A_1,\ldots,A_m} \otimes \mathbb{1}^{A_{m+1},\ldots,A_k} \otimes \mathbb{1}^B \Omega_k\Big) \tag{A.3}$$

$$= \mathrm{tr}\Big(\frac{1}{k!}\sum_{\pi\in S_k} O_P^{A_1,\ldots,A_m} \otimes \mathbb{1}^{A_{m+1},\ldots,A_k} \otimes \mathbb{1}^B \pi^\dagger \Omega_k \pi\Big) \tag{A.4}$$

$$= \mathrm{tr}\Big(\frac{1}{k!}\sum_{\pi\in S_k} \pi O_P^{A_1,\ldots,A_m} \otimes \mathbb{1}^{A_{m+1},\ldots,A_k} \otimes \mathbb{1}^B \pi^\dagger \Omega_k\Big) \tag{A.5}$$

$$= \mathrm{tr}\Big(O_{P_k} \otimes \mathbb{1}^B \Omega_k\Big). \tag{A.6}$$

□

Appendix B

Proofs for the SDP hierarchy

B.1 Convergence

Proof of Result 3.1. We consider a polynomial P of degree m. At level $k \geq m$ of the hierarchy, we obtain the optimal solution Ω_k, with $P_{\mathrm{LB}_k}(\rho) = \mathrm{tr}(O_P\, \Omega_m)$ where $\Omega_m = \mathrm{tr}_{m+1,\dots,k}(\Omega_k)$ is the m-copy marginal of Ω_k. By the de Finetti theorem for Bose-symmetric states [39, Thm. II.8], for every feasible Ω_k, there is an $\omega_m = \int |\phi\rangle\langle\phi|^{\otimes m}\, d\mu(\phi)$ such that

$$\|\Omega_m - \omega_m\|_1 \leq \epsilon \tag{B.1}$$

with $\epsilon = \frac{4Dm}{k}$. By Carathéodory's theorem, since the space is finite-dimensional, we can write $\omega_m = \sum_i p_i\, |\phi_i\rangle\langle\phi_i|^{\otimes m}$. Here, ω_m is associated to a decomposition $\{p_i, |\phi_i\rangle\}_i$ corresponding to a state σ and $\mathrm{tr}(O_P\, \omega_m) \geq \widehat{P}(\sigma)$. Since the partial trace is contractive, $\|\rho - \sigma\|_1 = \|\mathrm{tr}_{2,\dots,m}(\Omega_m - \omega_m)\|_1 \leq \epsilon$.

Then, by Hölder inequality (B.2), by definition of the convex roof (B.3) and from the continuity of the convex roof of a continuous function in finite dimensions (B.4) [56, Thm. 6.1],

$$\begin{aligned}
\mathrm{tr}(O_P \Omega_m) &\geq \mathrm{tr}(O_P \omega_m) - \epsilon \|O_P\|_\infty && \text{(B.2)}\\
&\geq \widehat{P}(\sigma) - \epsilon \|O_P\|_\infty && \text{(B.3)}\\
&\geq \widehat{P}(\rho) - \delta(\epsilon) - \epsilon \|O_P\|_\infty && \text{(B.4)}
\end{aligned}$$

with $\delta(\epsilon) \to 0$.

Hence, $\widehat{P}(\rho) \geq P_{\mathrm{LB}_k}(\rho) \geq \widehat{P}(\rho) - \delta(\epsilon) - \epsilon\|O_P\|_\infty$, and therefore $P_{\mathrm{LB}_k}(\rho) \to \widehat{P}(\rho)$ as $k \to \infty$. □

B.2 Monotonicity in the level

In the full SDP, at level $k+1$, we obtain the optimal solution Ω_{k+1}. Let O_k be the observable at level k and O_{k+1} be its lift to level $k+1$. Then, $P_{\mathrm{LB}_{k+1}} = \mathrm{tr}(O_{k+1}\Omega_{k+1}) = \mathrm{tr}(O_k \otimes \mathbb{1}\, \Omega_{k+1})$ according to Lemma A.1. At level k, $\mathrm{tr}_{A_{k+1}B_{k+1}}(\Omega_{k+1})$ is also in the feasible set – PPT across copies, permutation invariant, first marginal equal to ρ and PSD – with $\mathrm{tr}\Big(O_k\, \mathrm{tr}_{A_{k+1}B_{k+1}}(\Omega_{k+1})\Big) = P_{\mathrm{LB}_{k+1}}$. Then, $P_{\mathrm{LB}_{k+1}} \geq P_{\mathrm{LB}_k}$. We can use the same argument in k and m for the simplified SDP.

B.3 Simplified relaxation

Since $\Phi_{k,m}$ can be assumed invariant under permutations of $A_1B_1, \dots, A_kB_k$ and $A_{k+1}, \dots, A_m$, it suffices to impose partial transpose conditions on the tuples (i, j) where each tuple means

taking the partial transpose on the first i full-copy registers and the first j marginal registers. One such set of representatives is $\{(i,j) : 0 \le i \le k,\ 0 \le j < \frac{m-k}{2}\} \cup \{(i, \frac{m-k}{2}) : 0 \le i \le \lfloor k/2 \rfloor\}$ where the second set is only present for even $m-k$ and the tuple $(0,0)$ is excluded. By permutation invariance, adding any other PPT constraint would be redundant and not increase the bound. We also tried other sets of PPT constraints, such as enforcing them on $\mathrm{tr}_{B_1 \dots B_k}(\Phi_{k,m})$, which would have reduced the number of PPT constraints and their cost since they act on a smaller matrix, but the bounds we got were much worse.

Appendix C

Symmetry reductions and memory

Throughout this appendix, a single copy of $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$ is identified with $\mathbb{C}^D$, with orthonormal basis $\{|j\rangle\}_{j=0}^{D-1}$.

C.1 Restriction to the symmetric subspace

The symmetric subspace $\mathrm{Sym}^k(\mathbb{C}^D)$ has an orthonormal basis indexed by occupation vectors. The occupation vector denoted $\nu(i_1, \ldots, i_k) \in \mathbb{N}^D$ of a basis vector $|i_1, \ldots, i_k\rangle \in (\mathbb{C}^D)^{\otimes k}$ has entries ν_j equal to the number of times j appears in $(i_1, \ldots, i_k)$. For example, $\nu(0, 1, 2, 0, 0, 1, 2, 2) = (3, 2, 3)$ for $D = 3$, $k = 8$. We define the isometry $V_k : (\mathbb{C}^D)^{\otimes k} \to \mathrm{Sym}^k(\mathbb{C}^D)$ as

$$V_k = \sum_{i_1,\ldots,i_k=0}^{D-1} \binom{k}{\nu(i_1, \ldots, i_k)}^{-1/2} |\nu(i_1, \ldots, i_k)\rangle \langle i_1, \ldots, i_k| \,, \tag{C.1}$$

where $\binom{k}{\nu} = k!/\prod_j \nu_j!$ is the multinomial coefficient, i.e., the number of basis vectors with occupation vector ν. It has real entries and satisfies $V_k V_k^\dagger = \mathbb{1}_{\mathrm{Sym}^k}$ and $V_k^\dagger V_k = \Pi_{\mathrm{sym}}^k$. Since every feasible Ω_k is supported on the symmetric subspace, $\Omega_k = V_k^\dagger \Omega_k^{\mathrm{sym}} V_k$ with $\Omega_k^{\mathrm{sym}} = V_k \Omega_k V_k^\dagger$, a matrix of size $\binom{k+D-1}{k}$. The objective becomes $\mathrm{tr}\left(V_k (O_P \otimes \mathbb{1}^B) V_k^\dagger \, \Omega_k^{\mathrm{sym}}\right)$.

C.2 First marginal constraint

Here we wish to encode the condition $\mathrm{tr}_{A_2 B_2 \ldots A_k B_k}(\Omega_k) = \rho$. We define the occupation vector $\nu_\downarrow$ such that $|\nu_\downarrow\rangle \in \mathrm{Sym}^{k-1}(\mathbb{C}^D)$, and $|\nu_\downarrow + e_j\rangle \in \mathrm{Sym}^k(\mathbb{C}^D)$ has one more element $|j\rangle$, i.e., $(\nu_\downarrow + e_j)_j = (\nu_\downarrow)_j + 1$. Grouping the $\binom{k-1}{\nu_\downarrow}$ tuples $(i_2, \ldots, i_k)$ with the same occupation vector and using $\binom{k}{\nu_\downarrow + e_j} = \binom{k-1}{\nu_\downarrow} \frac{k}{\nu_{\downarrow,j}+1}$, we get

$$(\mathrm{tr}_{2,\ldots,k} \Omega_k)_{jj'} = \sum_{\nu_\downarrow} \binom{k-1}{\nu_\downarrow} \frac{\langle \nu_\downarrow + e_j | \, \Omega_k^{\mathrm{sym}} \, | \nu_\downarrow + e_{j'} \rangle}{\sqrt{\binom{k}{\nu_\downarrow + e_j}\binom{k}{\nu_\downarrow + e_{j'}}}} = \frac{1}{k} \mathrm{tr}\left(\hat{a}_{j'}^\dagger \hat{a}_j \, \Omega_k^{\mathrm{sym}}\right), \tag{C.2}$$

with the bosonic creation operators $\hat{a}_j^\dagger : \mathrm{Sym}^{k-1}(\mathbb{C}^D) \to \mathrm{Sym}^k(\mathbb{C}^D)$,

$$\hat{a}_j^\dagger = \sum_{\nu_\downarrow} \sqrt{\nu_{\downarrow,j} + 1} \; |\nu_\downarrow + e_j\rangle \langle \nu_\downarrow| \,. \tag{C.3}$$

Thus, the marginal constraint becomes the linear constraint $\frac{1}{k} \sum_{j,j'} \mathrm{tr}\left(\hat{a}_{j'}^\dagger \hat{a}_j \, \Omega_k^{\mathrm{sym}}\right) |j\rangle \langle j'| = \rho$.

C.3 Partial-transpose constraints

Thanks to the permutation symmetry, the partial transpose only depends on the size of the set of transposed copies, and since $\Omega_k^{T_{S^c}} = (\Omega_k^{T_S})^T$, a set and its complement give equivalent constraints. We therefore impose the PPT constraints on the first l copies for $l \in \{1, \ldots, \lfloor \frac{k}{2} \rfloor\}$. We define the isometry

$$W_l = (V_l \otimes V_{k-l}) V_k^\dagger : \ \mathrm{Sym}^k(\mathbb{C}^D) \to \mathrm{Sym}^l(\mathbb{C}^D) \otimes \mathrm{Sym}^{k-l}(\mathbb{C}^D), \tag{C.4}$$

so that $\Omega_k = (V_l \otimes V_{k-l})^\dagger \, W_l \Omega_k^{\mathrm{sym}} W_l^\dagger \, (V_l \otimes V_{k-l})$. Writing T_l also for the transpose of the factor $\mathrm{Sym}^l(\mathbb{C}^D)$ in the occupation basis, the partial transpose becomes

$$\Omega_k^{T_l} = \Big((V_l \otimes V_{k-l})^\dagger \, W_l \Omega_k^{\mathrm{sym}} W_l^\dagger \, (V_l \otimes V_{k-l}) \Big)^{T_l} \tag{C.5}$$

$$= \Big(V_l^T \otimes V_{k-l}^\dagger \Big) \, \Big(W_l \Omega_k^{\mathrm{sym}} W_l^\dagger \Big)^{T_l} \, (V_l^* \otimes V_{k-l}) \tag{C.6}$$

$$= (V_l \otimes V_{k-l})^\dagger \, \Big(W_l \Omega_k^{\mathrm{sym}} W_l^\dagger \Big)^{T_l} \, (V_l \otimes V_{k-l}), \tag{C.7}$$

since V_l is real, so that $V_l^T = V_l^\dagger$ and $V_l^* = V_l$. Conjugating back with $V_l \otimes V_{k-l}$, the condition becomes

$$\Omega_k^{T_l} \succeq 0 \iff \Big(W_l \, \Omega_k^{\mathrm{sym}} W_l^\dagger \Big)^{T_l} \succeq 0, \qquad 1 \le l \le \lfloor k/2 \rfloor. \tag{C.8}$$

This is the most expensive constraint of the SDP, since it encodes the state in the larger space $\mathrm{Sym}^l(\mathbb{C}^D) \otimes \mathrm{Sym}^{k-l}(\mathbb{C}^D)$ to support the partial transpose. While these constraints are not necessary for the hierarchy to be complete, they help convergence at finite k, as shown in Ref. [40].

C.4 Real variables

If ρ is real, the SDP can be restricted to real symmetric matrices. If Ω_k is feasible, so is its entrywise complex conjugate Ω_k^*, since conjugation commutes with partial traces, partial transposes and the real projector Π_{sym}^k, and $\rho^* = \rho$. Since the isotypic projectors are linear combinations of permutation matrices with real characters, O_P is real, and

$$\mathrm{tr}\Big((O_P^A \otimes \mathbb{1}^B) \, \Omega_k \Big) = \mathrm{tr}\Big((O_P^A \otimes \mathbb{1}^B) \, \Omega_k^* \Big) = \mathrm{tr}\Big((O_P^A \otimes \mathbb{1}^B) \, \tfrac{\Omega_k + \Omega_k^*}{2} \Big). \tag{C.9}$$

By convexity, $\frac{1}{2}(\Omega_k + \Omega_k^*)$ is feasible, real and has the same objective value, so the optimum is attained by a real matrix. For the true ω_k, this real matrix is $\frac{1}{2} \sum_i p_i \left(|\psi_i\rangle\langle\psi_i|^{\otimes k} + |\psi_i^*\rangle\langle\psi_i^*|^{\otimes k} \right)$, which corresponds to the decomposition $\{\frac{p_i}{2}, |\psi_i\rangle\} \cup \{\frac{p_i}{2}, |\psi_i^*\rangle\}$ of ρ. The same argument applies to $\Phi_{k,m}$ in the simplified relaxation.

C.5 Projection onto the support of ρ

Let $r = \mathrm{rank}\, \rho$, and let $V_\rho : \mathbb{C}^D \to \mathbb{C}^r$ be the isometry onto the support of ρ, given by the eigenvectors of ρ with nonzero eigenvalue, so that $V_\rho V_\rho^\dagger = \mathbb{1}_r$ and $V_\rho^\dagger V_\rho = \Pi_\rho$, the projector onto the support of ρ. Every feasible Ω_k is supported on $(\mathrm{supp}\, \rho)^{\otimes k}$: since

$$\mathrm{tr}\left(((\mathbb{1} - \Pi_\rho) \otimes \mathbb{1}) \, \Omega_k \right) = \mathrm{tr}\left((\mathbb{1} - \Pi_\rho) \rho \right) = 0 \tag{C.10}$$

and $\Omega_k \succeq 0$, we have $((\mathbb{1} - \Pi_\rho) \otimes \mathbb{1})\,\Omega_k = 0$, and by permutation symmetry the same holds on every copy. Hence $\Omega_k = \Pi_\rho^{\otimes k} \Omega_k \Pi_\rho^{\otimes k}$, and

$$\operatorname{tr}\left((O_P^A \otimes \mathbb{1}^B)\,\Omega_k\right) = \operatorname{tr}\left(\left(V_\rho^{\otimes k}(O_P^A \otimes \mathbb{1}^B)V_\rho^{\dagger\otimes k}\right)\left(V_\rho^{\otimes k}\Omega_k V_\rho^{\dagger\otimes k}\right)\right). \tag{C.11}$$

We can therefore equivalently optimize over $\Omega_k^\rho = V_\rho^{\otimes k}\Omega_k V_\rho^{\dagger\otimes k}$ on $(\mathbb{C}^r)^{\otimes k}$ with marginal $V_\rho \rho V_\rho^\dagger$. Bose symmetry is preserved, since $V_\rho^{\otimes k}$ commutes with permutations, and the partial-transpose constraints are equivalent, since $(\Omega_k^\rho)^{T_l}$ and $\Omega_k^{T_l}$ are related in both directions by congruences with V_ρ and V_ρ^*. Combined with the symmetric subspace, the variable has size $\binom{k+r-1}{k}$. If ρ is real, V_ρ can be chosen real and the real-variable reduction still applies.

The same argument applies to the simplified relaxation: by the permutation invariance of $\operatorname{tr}_B \Phi_{k,m}$, every A register has reduced state $\rho_A = \operatorname{tr}_B(\rho)$, and therefore all A registers can be encoded on the support of ρ_A.

C.6 Memory cost

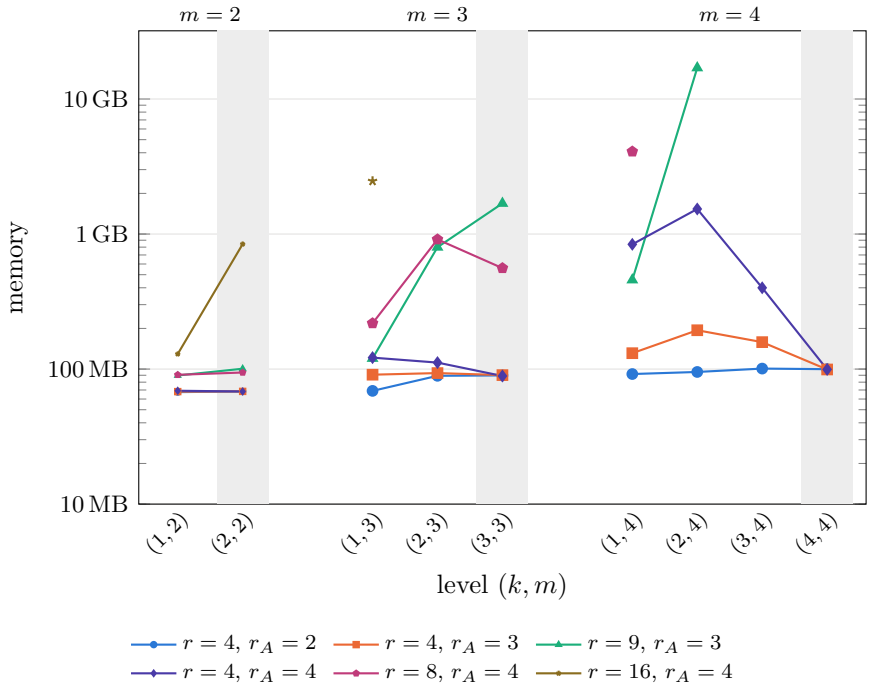


Figure C.1: Memory needed by the SDP (3.13) with the QICS solver [57] at different levels for random states ρ with $r = \operatorname{rank}(\rho)$ and $r_A = \operatorname{rank}(\operatorname{tr}_B(\rho))$. When $k = m$, the full SDP (3.11) is used.

Appendix D

Extraction of decompositions

D.1 Real tensor representation

We take the optimal solution Ω_k of Eq. (3.11). We work on the support of ρ as in Appendix C.5: with $r = \operatorname{rank}\rho$ and the isometry $V_\rho : \operatorname{supp}\rho \to \mathbb{C}^r$, we consider $\Omega_\rho = V_\rho^{\otimes k} \Omega_k V_\rho^{\dagger\otimes k}$.

To work with real tensors, let $\{G_a\}_{a=0}^{r^2-1}$ be a Hilbert–Schmidt orthonormal basis of Hermitian operators on $\mathbb{C}^r$, consisting of the normalized identity $G_0 = \mathbb{1}/\sqrt{r}$ and the generalized Gell-Mann matrices. Every pure state can be written as $|\psi\rangle\langle\psi| = \sum_a x_a G_a$ with a real vector x satisfying $x_0 = 1/\sqrt{r}$ and $\|x\|_2 = 1$. We define the real symmetric order-k tensor $\Omega^G_{a_1\ldots a_k} = \operatorname{tr}\left((G_{a_1} \otimes \cdots \otimes G_{a_k})\,\Omega_\rho\right)$, which for the true ω_k equals $\sum_i p_i\, x^{(i)\otimes k}$. We run PARAFAC (alternating least squares) on Ω^G with nonnegative weights, obtaining terms $\{p_i, v_i\}$; we normalize each factor such that $v_{i,0} = 1/\sqrt{r}$, absorbing the scale into p_i, and recover the pure states $|\psi_i\rangle$ as the top eigenvectors of $\sum_a v_{i,a} G_a$. Finally, we repair the decomposition $\{p_i, |\psi_i\rangle\}$ as described in Sec. 4.2.

The number of terms n fitted by PARAFAC is the rank of Ω_ρ, which equals the number of terms when the $|\psi_i\rangle\langle\psi_i|^{\otimes k}$ are linearly independent. Any decomposition of ρ has at least r terms, otherwise the repair fails. Conversely, every decomposition of ρ consists of states on $\operatorname{supp}\rho \cong \mathbb{C}^r$, whose state space has dimension $r^2 - 1$, and therefore by Carathéodory's theorem the convex roof of a continuous function is attained by a decomposition with at most r^2 terms [12, Lemma A-2].

D.2 Rank reduction

Before the extraction, it helps to reduce the rank of the SDP solution. We solve a second SDP, minimizing $\operatorname{tr}(R_3 \otimes \mathbb{1}\,\Omega_k)$ for a random Hermitian R_3 acting on the first 3 copies, over the same feasible set with the additional constraint $\operatorname{tr}\left((O_P \otimes \mathbb{1}^B)\Omega_k\right) \le P_{\mathrm{LB}} + \delta_k$ for a small $\delta_k \to 0$ with $\delta_k \ge \widehat{P}(\rho) - P_{\mathrm{LB}_k}$. This selects an extreme point of the near-optimal face, which generically has low rank, instead of a mixture of several optimal solutions.

D.3 Convergence of the extraction

Given a state $\omega_3 = \sum_i p_i\, |\psi_i\rangle\langle\psi_i|^{\otimes 3}$, we can use Jennrich's algorithm to recover uniquely the decomposition $\{p_i, |\psi_i\rangle\}_i$ when the $|\psi_i\rangle\langle\psi_i|$ are linearly independent. Therefore, we need to select a low-rank extreme point $\Omega_3 = \operatorname{tr}_{A_4 B_4 \ldots A_k B_k}(\Omega_k)$. For this we apply the rank reduction step of Sec. D.2.

Now we need the following observation.

Observation D.1. *A decomposition $\{q_i, |\psi_i\rangle\langle\psi_i|\}$ of ρ is extremal in the set of optimal decompositions of ρ with respect to a polynomial P only if the $|\psi_i\rangle\langle\psi_i|$ are linearly independent projectors.*

Proof. We will do so by contradiction as follows. Given an optimal decomposition $\{q_i, |\psi_i\rangle\langle\psi_i|\}$ of ρ, if the projectors $|\psi_i\rangle\langle\psi_i|$ are linearly dependent, there exists a nonzero real vector c such that $\sum_i c_i |\psi_i\rangle\langle\psi_i| = 0$ with $\sum_i c_i = 0$. Assuming that this is the case, for a small $0 < t \leq \min_i \frac{q_i}{|c_i|}$, let us define two other decompositions $\{q_i + tc_i, |\psi_i\rangle\langle\psi_i|\}$ and $\{q_i - tc_i, |\psi_i\rangle\langle\psi_i|\}$ such that their midpoint is the decomposition $\{q_i, |\psi_i\rangle\langle\psi_i|\}$. Since $\sum_i (q_i + tc_i)P(\psi_i) \geq \widehat{P}(\rho) = \sum_i q_i P(\psi_i)$ and $\sum_i (q_i - tc_i)P(\psi_i) \geq \widehat{P}(\rho)$, then $\sum_i (q_i + tc_i)P(\psi_i) = \sum_i (q_i - tc_i)P(\psi_i) = \widehat{P}(\rho)$. These decompositions are also optimal and their midpoint is the decomposition extracted $\{q_i, |\psi_i\rangle\langle\psi_i|\}$, which proves that it is not an extreme point. Therefore, any extreme point of the convex set of optimal decompositions must have linearly independent projectors. □

Proof of Result 4.1. For almost every R, the minimizer ω_3^* of $\mathrm{tr}(R\,\omega_3)$ among the ω_3 of optimal decompositions is unique. The decomposition of ρ giving ω_3^* has linearly independent projectors. Using the same argument shown above, if it has linearly dependent terms, we can define two decompositions whose associated ω_3 is optimal with midpoint ω_3^*. By uniqueness, both decompositions give ω_3^*. We can simply increase t until a weight vanishes and repeat to end up at a decomposition into linearly independent terms. Therefore, ω_3^* decomposes into linearly independent terms.

As $k \to \infty$ the rank-reduced Ω_3 converges to $\omega_3(\mu_e) = \sum_i p_i |\psi_i\rangle\langle\psi_i|^{\otimes 3}$, where μ_e is the extreme optimal decomposition singled out by R, since $\delta_k \geq \widehat{P}(\rho) - P_{\mathrm{LB}_k}$ leaves $\omega_3(\mu_e)$ feasible during the rank reduction step. By Appendix B.1, every limit point as $k \to \infty$ corresponds to the state ω_3 of an optimal decomposition, and for almost every R the minimizer of $\mathrm{tr}(R\,\cdot)$ on this set is unique and has linearly independent projectors. Then, Jennrich's algorithm is continuous at $\omega_3(\mu_e)$, and therefore applied to Ω_3 it gives a decomposition of a state σ that is ϵ-close to ρ with $\epsilon \to 0$ as $k \to \infty$, which after the repair step has value converging to $\widehat{P}(\rho)$. □

This argument does not extend to LOCC since the optimal decompositions do not form a face of the set of decompositions of ρ, and the extreme points may not have linearly independent projectors, in which case we cannot use Jennrich's algorithm.

Appendix E

Linear program for non-polynomial functions

E.1 Construction of the LP and of the single SDP

We optimize over symmetric polynomials of degree at most k. Since polynomials of lower degree can be lifted to degree k (Lemma A.1), we parametrize them as $P(\alpha) = \sum_{\lambda \vdash k} c_\lambda s_\lambda(\alpha)$, and the LP variables are the coefficients $\vec{c} = (c_\lambda)_\lambda$. Given the function F, we require $P(\alpha) \leq F(\alpha)$ for all spectra α, and enforce this constraint discretely on a grid of spectra $\{\alpha^g\}_g$, where the values $s_\lambda(\alpha^g)$ and $F(\alpha^g)$ are precomputed. By symmetry, it suffices to take grid points with $\alpha_{g,1} \geq \alpha_{g,2} \geq \cdots \geq \alpha_{g,d_A} \geq 0$ and $\sum_i \alpha_{g,i} = 1$.

Writing $\mu_\lambda(\Phi) = \operatorname{tr}\left((\frac{1}{f^\lambda}\Pi^\lambda \otimes \mathbb{1}^B)\Phi\right)$, the objective is $\operatorname{tr}\left((O_P \otimes \mathbb{1}^B)\Omega_k\right) = \sum_\lambda c_\lambda \mu_\lambda(\Omega_k)$, and therefore the bound reads

$$F_{\mathrm{LB}_k} = \max_{\vec{c}} \min_{\Omega_k} \sum_\lambda c_\lambda \mu_\lambda(\Omega_k) \quad \text{s.t.} \quad \sum_\lambda c_\lambda s_\lambda(\alpha^g) \leq F(\alpha^g) \ \ \forall g, \tag{E.1}$$

with Ω_k ranging over the feasible set of Eq. (3.11). Since the objective is bilinear, the set of feasible Ω_k is convex and compact, and the set of valid $\vec{c}$ is convex, we can exchange the maximum and the minimum [52]. Then, for fixed Ω_k, the inner maximization is an LP whose dual is

$$\min_{y \geq 0} \sum_g y_g F(\alpha^g) \quad \text{s.t.} \quad \sum_g y_g \, s_\lambda(\alpha^g) = \mu_\lambda(\Omega_k) \ \ \forall \lambda. \tag{E.2}$$

The primal is always feasible, since the constant polynomial $P = \min_g F(\alpha^g)$ is admissible, so strong duality holds. We merge the LP and the SDP to obtain a single SDP,

$$\begin{aligned} F_{\mathrm{LB}_k} = \min_{\Omega_k, y} & \sum_g y_g F(\alpha^g) \\ \text{s.t. } & \Omega_k \text{ feasible for Eq. (3.11)}, \quad y \geq 0, \\ & \sum_g y_g \, s_\lambda(\alpha^g) = \operatorname{tr}\left((\tfrac{1}{f^\lambda}\Pi^\lambda \otimes \mathbb{1}^B)\,\Omega_k\right) \quad \forall \lambda \vdash k. \end{aligned} \tag{E.3}$$

The coefficients c_λ of the optimal polynomial are recovered as the dual variables of the equality constraints. Since the grid is discrete, the optimal polynomial P^* may exceed F between grid points. We therefore report $F_{\mathrm{LB}_k} - \epsilon$, where $\epsilon = \max_\alpha (P^*(\alpha) - F(\alpha))$ is estimated from the polynomial returned by the solver. This is a valid lower bound, since $P^* - \epsilon \leq F$ on all spectra.

E.2 Proof of convergence

Here we prove Result 5.1 for the bound $\widetilde{F}_{\mathrm{LB}_k}$ of Eq. (5.1), in which $P \leq F$ is imposed on all spectra. Monotonicity follows naturally from the facts that admissible polynomials stay admissible at higher level by lifting (Lemma A.1), and that the SDP is monotone (Appendix B.2).

By the Weierstrass approximation theorem, for any continuous function F' and any $\eta > 0$, we can find a polynomial P' such that $|F'(\alpha) - P'(\alpha)| < \eta$ for all spectra α. Then, by averaging over permutations of the variables, for any continuous symmetric function F, we can find a symmetric polynomial P such that $|F(\alpha) - P(\alpha)| < \eta$ for all spectra α. Shifting P by η, for any F and any $\eta > 0$, there exists a symmetric polynomial P such that $P(\alpha) < F(\alpha)$ and $F(\alpha) - P(\alpha) < 2\eta$ for all spectra α. This P is admissible for $\deg P \leq k$, and therefore by Result 3.1, $\lim_{k\to\infty} \widetilde{F}_{\mathrm{LB}_k}(\rho) \geq \lim_{k\to\infty} P_{\mathrm{LB}_k}(\rho) = \widehat{P}(\rho) \geq \widehat{F}(\rho) - 2\eta$. Conversely, $P \leq F$ implies $\widehat{P}(\rho) \leq \widehat{F}(\rho)$ for every admissible P, and therefore $\widetilde{F}_{\mathrm{LB}_k}(\rho) \leq \widehat{F}(\rho)$. Since η is arbitrary,

$$\lim_{k\to\infty} \widetilde{F}_{\mathrm{LB}_k}(\rho) = \widehat{F}(\rho). \tag{E.4}$$

With a finite grid, the set of admissible polynomials is larger, and therefore $F_{\mathrm{LB}_k} \geq \widetilde{F}_{\mathrm{LB}_k}$, and $F_{\mathrm{LB}_k} - \epsilon \leq \widehat{F}(\rho)$, where ϵ is the grid error of the returned polynomial. The uncorrected value F_{LB_k} is nondecreasing in the level by the same argument as above, but the corrected value $F_{\mathrm{LB}_k} - \epsilon$ need not be, since the grid error can change between levels. In our numerical results, we observe that increasing the level yields tighter bounds.

Appendix F

LOCC feasibility SDP

F.1 Construction of the SDP

Let $|\psi\rangle$ be the initial pure state and ρ the target mixed state. By Ref. [51], $|\psi\rangle$ can be converted into ρ by LOCC if and only if there is a decomposition $\{p_i, |\psi_i\rangle\}$ of ρ such that

$$\sum_i p_i E_j(\psi_i) \leq E_j(\psi) \qquad \text{for all } j = 1, \ldots, d_A - 1. \tag{F.1}$$

Since all these inequalities must hold for the same decomposition, testing them one monotone at a time is not sufficient. Instead, we combine the LP construction of Appendix E.1 for all the Vidal monotones into a single SDP.

For each j, let $\mathcal{P}_j^G$ be the set of symmetric polynomials P_j of degree at most k that satisfy $P_j(\alpha^g) \leq E_j(\alpha^g)$ on all grid points, and let

$$\epsilon_j(P_j) = \max_\alpha \left(P_j(\alpha) - E_j(\alpha)\right) \tag{F.2}$$

be the maximal violation of P_j between grid points. If a decomposition $\{p_i, |\psi_i\rangle\}$ satisfies Eq. (F.1), then its operator ω_k is feasible for Eq. (3.11) and, for every $P_j \in \mathcal{P}_j^G$,

$$\operatorname{tr}\left((O_{P_j} \otimes \mathbb{1}^B)\,\omega_k\right) = \sum_i p_i P_j(\alpha_i) \leq \sum_i p_i E_j(\alpha_i) + \epsilon_j(P_j) \leq E_j(\psi) + \epsilon_j(P_j), \tag{F.3}$$

where α_i is the entanglement spectrum of $|\psi_i\rangle$. Therefore, if a certain decomposition can be transformed from $|\psi\rangle$ with LOCC, then it can violate the j-th inequality by at most the grid error $\epsilon_j(P_j)$ for each polynomial.

With the moments $\mu_\lambda(\Phi) = \operatorname{tr}\left((\frac{1}{f^\lambda}\Pi^\lambda \otimes \mathbb{1}^B)\Phi\right)$ of Appendix E.1, the optimization of Eq. (6.3) is computed by the SDP

$$\begin{aligned} s_k = \min_{\Omega_k,\, y^{(j)},\, s} \quad & s \\ \text{s.t. } & \Omega_k \text{ feasible for Eq. (3.11)}, \quad y^{(j)} \geq 0 \;\; \forall j, \\ & \sum_g y_g^{(j)}\, s_\lambda(\alpha^g) = \mu_\lambda(\Omega_k) \quad \forall \lambda \vdash k,\; \forall j, \\ & \sum_g y_g^{(j)}\, E_j(\alpha^g) - E_j(\psi) \leq s \quad \forall j. \end{aligned} \tag{F.4}$$

For fixed Ω_k, the minimization over each $y^{(j)}$ is the dual LP of Appendix E.1 with $F = E_j$, and therefore by strong duality

$$s_k = \min_{\Omega_k} \max_j \max_{P_j \in \mathcal{P}_j^G} \left(\operatorname{tr}\left((O_{P_j} \otimes \mathbb{1}^B)\Omega_k\right) - E_j(\psi)\right). \tag{F.5}$$

Thus s_k is the smallest worst-case violation of Eq. (F.1) over the relaxations of decompositions of ρ, where each violation is measured with the best grid-admissible polynomial. For a single monotone, $s_k = F_{\mathrm{LB}_k} - E_j(\psi)$ with $F = E_j$.

Let t be a probability vector with $t_j \geq 0$ and $\sum_j t_j = 1$. Then,

$$s_k = \min_{\Omega_k} \max_{t} \max_{P_j \in \mathcal{P}_j^G} \sum_j t_j \Big(\operatorname{tr}\left((O_{P_j} \otimes \mathbb{1}^B)\Omega_k\right) - E_j(\psi)\Big). \tag{F.6}$$

With the substitution $Q_j = t_j P_j$, the constraint $P_j \in \mathcal{P}_j^G$ becomes $Q_j(\alpha^g) \leq t_j E_j(\alpha^g)$ for all g, and the objective is linear in Ω_k and in (t, Q_j). The feasible set of Ω_k is convex and compact, and the set of (t, Q_j) is convex. Again, here the minimum and the maximum can be exchanged by Ref. [52]:

$$\begin{aligned} s_k &= \min_{\Omega_k} \max_{t, Q_j} \sum_j \Big(\operatorname{tr}\left((O_{Q_j} \otimes \mathbb{1}^B)\Omega_k\right) - t_j E_j(\psi)\Big) \\ &= \max_{t, Q_j} \min_{\Omega_k} \sum_j \Big(\operatorname{tr}\left((O_{Q_j} \otimes \mathbb{1}^B)\Omega_k\right) - t_j E_j(\psi)\Big). \end{aligned} \tag{F.7}$$

Let t^* and Q_j^* be the optimal vector t and optimal polynomials, which are given by the dual variables of the SDP (F.4); for $t_j^* > 0$, $Q_j^* = t_j^* P_j^*$ with $P_j^* \in \mathcal{P}_j^G$. By Eq. (F.7), they satisfy

$$\sum_j \Big(\operatorname{tr}\left((O_{Q_j^*} \otimes \mathbb{1}^B)\Omega_k\right) - t_j^* E_j(\psi)\Big) \geq s_k \tag{F.8}$$

for every feasible Ω_k.

Let $\epsilon_j = \max_\alpha \left(Q_j^*(\alpha) - t_j^* E_j(\alpha)\right)$, which equals $t_j^* \, \epsilon_j(P_j^*)$ for $t_j^* > 0$. If $|\psi\rangle$ can be converted into ρ by LOCC, some decomposition satisfies Eq. (F.1), and its ω_k is feasible. As in Eq. (F.3), $\operatorname{tr}\left((O_{Q_j^*} \otimes \mathbb{1}^B)\,\omega_k\right) = \sum_i p_i Q_j^*(\alpha_i) \leq t_j^* E_j(\psi) + \epsilon_j$, and therefore Eq. (F.8) gives

$$s_k \leq \sum_j \Big(\operatorname{tr}\left((O_{Q_j^*} \otimes \mathbb{1}^B)\,\omega_k\right) - t_j^* E_j(\psi)\Big) \leq \sum_j \epsilon_j. \tag{F.9}$$

Thus, if $s_k > \sum_j \epsilon_j$, $|\psi\rangle$ cannot be converted into ρ by LOCC. If the condition does not hold, the test does not tell us anything, in which case, for $k \geq 3$, we can extract the decomposition from Ω_k and repair it as in Chapter 4, and if the criterion Eq. (F.1) holds, the state can be converted into ρ by LOCC. In this procedure, the grid errors ϵ_j need to be computed in a postprocessing step from the polynomials returned by the solver.

F.2 Detection at finite level

We impose $P_j \leq E_j$ on all spectra. Consider the decomposition $\{p_i, |\varphi_i\rangle\langle\varphi_i|\}$ of ρ that minimizes

$$s^* = \max_t \sum_j t_j \left(\left(\sum_i p_i E_j(\varphi_i) \right) - E_j(\psi) \right) \tag{F.10}$$

over probability vectors t. If $|\psi\rangle \overset{\text{LOCC}}{\nrightarrow} \rho$, at least one Vidal monotone E_j will be exceeded by the decomposition and setting $t_j = 1$, we get that $s^* > 0$. We know $\widetilde{s}_k \leq s^*$ since we are relaxing the set of feasible ω_k. By Appendix B.1, as $k \to \infty$, the few-copy marginals of Ω_k approach mixtures $\sum q_i \, |\phi_i\rangle\langle\phi_i|^{\otimes n}$ with $\{q_i, |\phi_i\rangle\langle\phi_i|\}$ a decomposition of ρ, and since by Appendix E.2 each E_j can be approximated to arbitrary precision from below, $\widetilde{s}_k \to s^*$ as $k \to \infty$ and every non-convertible pair is eventually detected.

Appendix G

Further numerical results

In this appendix, we provide additional numerical results for the entanglement of formation (Fig. G.1), the linear entropy (Fig. G.2) and the Vidal monotones (Fig. G.3). We also show how the LP adapts to different states by choosing different lower-bounding polynomials, which are tangent to the reference function at different points of the spectrum (Fig. G.4).

All the SDPs are solved with the interior-point conic solver QICS [57], with its default tolerances. The PPT entangled states from the figures come from the library `bound-entangled` [55] and the NPT ones are built with `toqito` [58]. For the Horodecki 3×3 and 2×4 states we take the parameter $a = 0.5$ in Figs. G.1, G.2 and G.3. The parameters of the NPT states are the ones given in the caption of Fig. G.4.

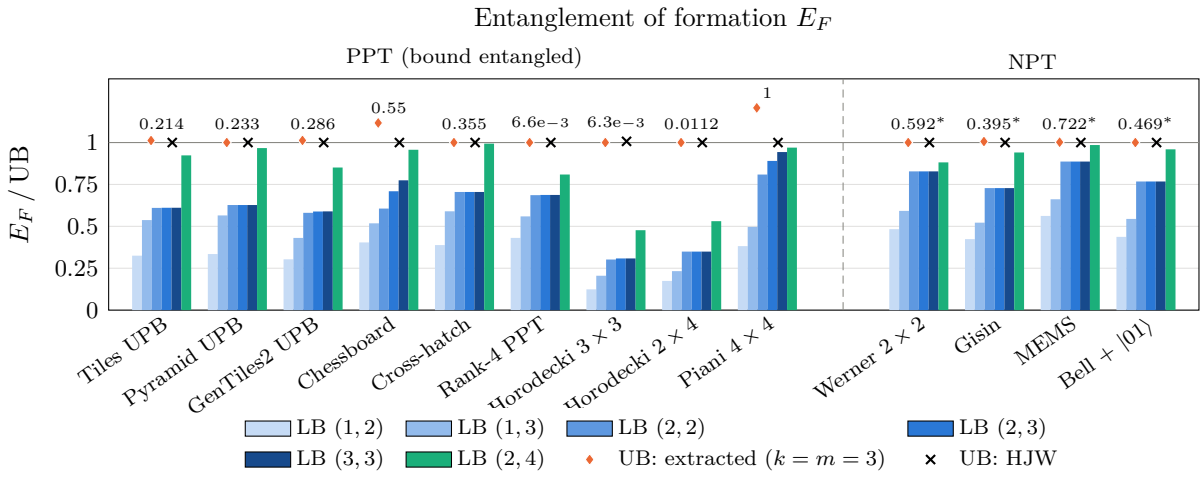


Figure G.1: Lower bounds on the entanglement of formation E_F for different PPT and NPT states, from the SDP with embedded LP. Bars: lower bounds from the simplified hierarchy at the levels (k, m) shown in the legend. Markers: upper bounds from the decomposition extracted from the SDP solution at $k = m = 3$ (orange diamonds) and from HJW gradient descent over decompositions (black crosses). All values are divided by the best upper bound UB, printed above each state. Here, UB is the exact convex roof for two qubits (marked *), and the lowest of the two upper bounds otherwise. Notice that the gap between upper and lower bounds decreases with the hierarchy level.

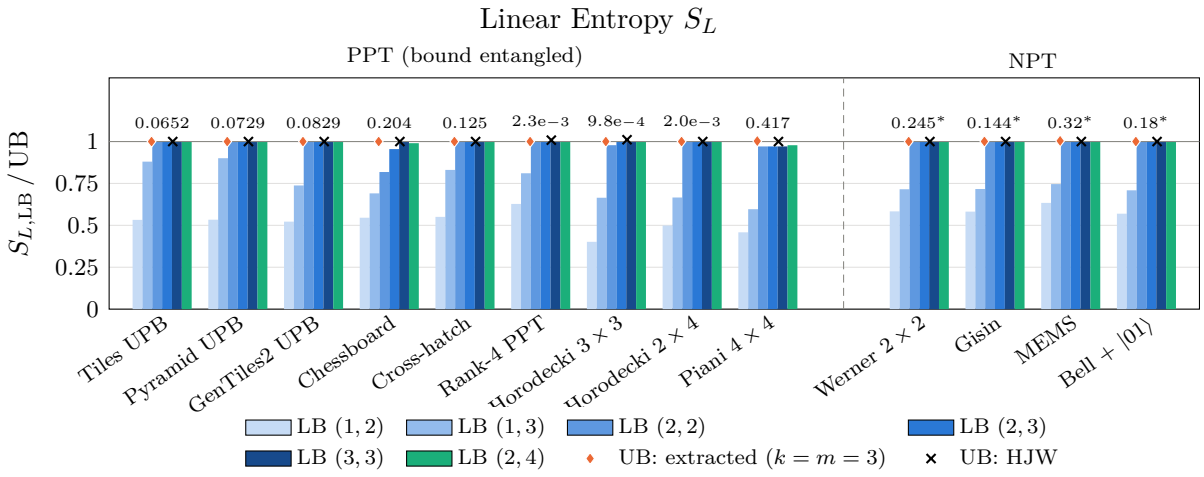


Figure G.2: Lower bounds on the linear entropy for different PPT and NPT states. Bars: lower bounds from the simplified hierarchy at the levels (k, m) shown in the legend. Markers: upper bounds from the decomposition extracted from the SDP solution at $k = m = 3$ (orange diamonds) and from HJW gradient descent over decompositions (black crosses). All values are divided by the best upper bound UB, printed above each state. Here, UB is the exact convex roof for two qubits (marked *), and the lowest of the two upper bounds otherwise.

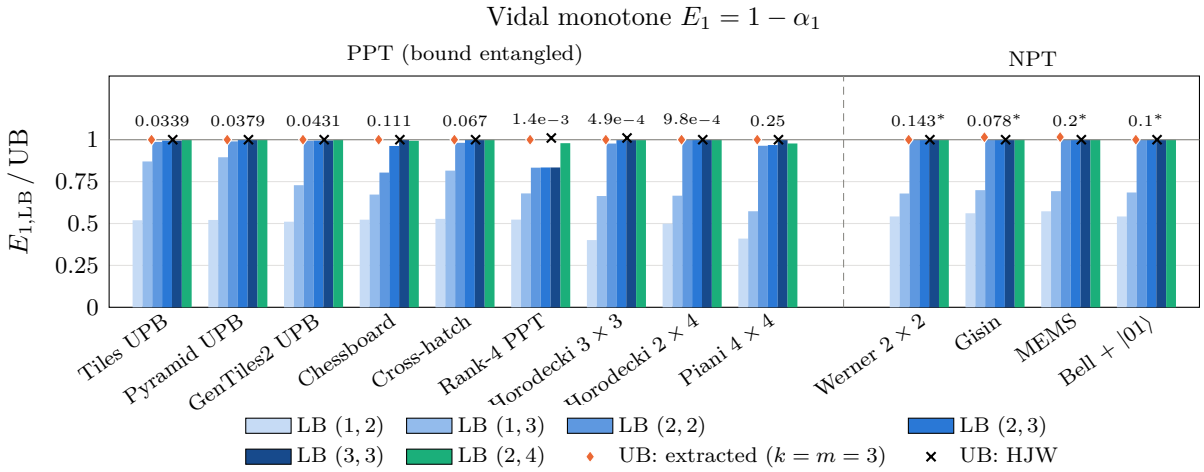


Figure G.3: Lower bounds on the Vidal monotone E_1 for different PPT and NPT states, from the SDP with embedded LP. Bars: lower bounds from the simplified hierarchy at the levels (k, m) shown in the legend. Markers: upper bounds from the decomposition extracted from the SDP solution at $k = m = 3$ (orange diamonds) and from HJW gradient descent over decompositions (black crosses). All values are divided by the best upper bound UB, printed above each state. Here, UB is the exact convex roof for two qubits (marked *), and the lowest of the two upper bounds otherwise.

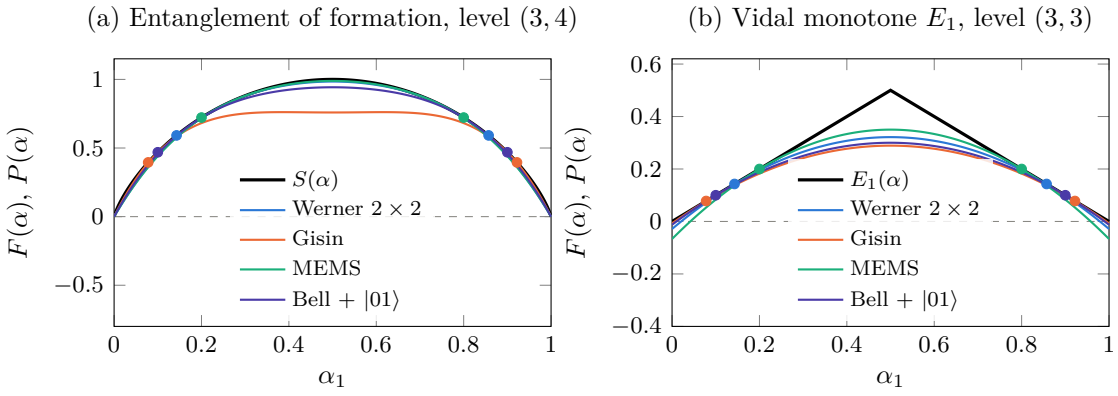


Figure G.4: Polynomials $P \leq F$ extracted from the SDP with embedded LP for different 2-qubit states. For each state, this is the polynomial that yields the best bound on the convex roof on the grid, at level $(3, 4)$ for E_F (a) and at level $(3, 3)$ for E_1 (b). Here we show that the LP is adaptive depending on the state and there is not a single polynomial that yields the best bound on the convex roof for all states at the same level. The states used are { Werner 2×2: singlet weight $p = 0.8$; Gisin: $\lambda = 0.9$, $\theta = \pi/8$; MEMS: $C = 0.8$; Bell + $|01\rangle$: $p = 0.6$ }. The polynomials are shown as a function of the entanglement spectrum $\alpha = (\alpha_1, \alpha_2) = (\alpha_1, 1 - \alpha_1)$.

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